AN ADAPTIVE AND TEMPORALLY CONSISTENT LANE DETECTION FRAMEWORK FOR ADVANCED DRIVER ASSISTANCE SYSTEMS

A PROJECT REPORT

Submitted by

AVANTHIKA S 810022106022

VEERAPRADHA M 810022106027

LOKESWARI R 810022106029

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TIRUCHIRAPPALLI - 620 024

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ANNA UNIVERSITY : CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report "**AN ADAPTIVE AND TEMPORALLY CONSISTENT LANE DETECTION FRAMEWORK FOR ADAVNCED DRIVER ASSISTANCE SYSTEM**" is the bonafide work of **AVANTHIKA S (810022106022), VEERAPRADHA M (810022106027), LOKESWARI R (810022106029)**, who carried out the project work under my supervision.

Signature

Dr.P.RAMADEVI,

Head of the Department,

Associate Professor,

Department of ECE,

University College of Engineering,

Anna University – BIT Campus,

Tiruchirappalli - 620024.

Signature

Dr. R. SOWMYAL AKSHMI

Supervisor,

Assistant Professor (Sl.Gr)

Department of ECE,

University College of Engineering,

Anna University – BIT Campus ,

Tiruchirappalli - 620024.

Submitted for project viva-voce Examination held on.....

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

We hereby declare that the work entitled "**AN ADAPTIVE AND TEMPORALLY CONSISTENT LANE DETECTION FRAMEWORK FOR ADAVNCED DRIVER ASSISTANCE SYSTEM**" is submitted in partial fulfillment of the requirement for the award of the degree in B.E., Electronics and Communication Engineering, is a record of my own work carried out by me during the Academic year 2025 - 2026 under the Supervision and guidance of Dr. R. Sowmyalakshmi, Assistant Professor (Sl.Gr), Department of Electronics and Communication Engineering, **University College of Engineering - BIT Campus, Anna University, Tiruchirappalli - 620024**. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any other degree, either in this or any other University.

AVANTHIKA S **810022106022**

VEERAPRADHA M **810022106027**

LOKESWARI R **810022106029**

I certify that the above declaration made by the candidate is true.

SIGNATURE OF SUPERVISOR

Dr. R. SOWMYALAKSHMI

Assistant Professor (Sl.Gr)/ ECE

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ABSTRACT

Lane detection is an essential component of Advanced Driver Assistance Systems (ADAS), enabling vehicles to identify road boundaries and maintain safe driving trajectories. However, many existing methods process images independently, resulting in unstable performance under challenging conditions such as shadows, illumination variations, and curved roads. This paper presents an adaptive lane detection framework that enhances both accuracy and temporal stability using image-based inputs. Unlike conventional approaches, the proposed system maintains consistency across consecutive images to reduce detection fluctuations. The method combines Sliding Discrete Cosine Transform (SDCT) for effective edge extraction with Random Sample Consensus (RANSAC) for robust lane modeling. The processing pipeline includes grayscale conversion and Region of Interest (ROI) selection to focus on relevant road areas, followed by SDCT-based edge detection and blob analysis to group edge pixels into meaningful lane segments. RANSAC is then applied to fit lane models while removing outliers, enabling accurate detection of both straight and curved lanes. A temporal consistency mechanism further improves detection continuity across inputs.

The system is implemented using Python with OpenCV and FPGA using VHDL for efficient execution. Experimental results demonstrate an accuracy of 96–99%, making the approach suitable for real-time ADAS applications without relying on computationally intensive deep learning techniques.

சுருக்கம்

சாலைப் பாதை கண்டறிதல் என்பது ADAS அமைப்பின் முக்கிய அங்கமாகும்; இது வாகனங்களுக்கு சாலை எல்லைகளை கண்டறிந்து பாதுகாப்பான ஓட்டத்தை வழங்க உதவுகிறது. ஆனால், தற்போதைய முறைகள் ஒவ்வொரு படத்தையும் தனித்தனியாக செ யலாக்குவதால் நிழல் மற்றும் ஒளி மாற்றங்களில் நிலைத்தன்மை குறைகிறது. இதைத் தீர்க்க, இந்த ஆய்வு தொடர்ச்சியான படங்களில் துல்லியம் மற்றும் நிலைத்தன்மையை மேம்படுத்தும் ஒரு தகவமைப்பு முறையை முன்வைக்கிறது. இதில் SDCT மூலம் edge detection மற்றும் RANSAC மூலம் lane modeling செய்யப்படுகிறது; மேலும் grayscale மாற்றம், ROI தேர்வு, blob analysis ஆகியவற்றின் மூலம் lane பகுதிகள் கண்டறியப்படுகின்றன. Temporal consistency மூலம் detection நிலைத்தன்மை அதிகரிக்கப்படுகிறது.

இந்த முறை Python, OpenCV மற்றும் FPGA (VHDL) மூலம் செயல்படுத்தப்பட்டு, 96–99% துல்லியத்துடன் real-time ADAS பயன்பாடுகளுக்கு ஏற்றதாக உள்ளது.

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LIST OF ABBREVIATION

ABBREVIATION	EXPANSION
ADAS	Advanced Driver Assistance Systems
CLAHE	Contrast Limited Adaptive Histogram Equalization
DCT	Discrete Cosine Transform
FPGA	Field Programmable Gate Array
GB	Gaussian Blur
GPU	Graphics Processing Unit
HLS	Hue, Light, Saturation
IDCT	Inverse Discrete Cosine Transform
ITS	Intelligent Transport System
LiDAR	Light Detection and Ranging
OpenCV	Open Source Computer Vision Library
PHT	Probabilistic Hough Transform
RAM	Random Access Memory
RADAR	Radio Detection and Ranging
RGB	Red, Green, Blue
ROI	Region of Interest
RANSAC	Random Sample Consensus
SDCT	Sliding Discrete Cosine Transform
VHDL	Very High Speed Integrated Circuit Hardware Description Language

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVE OF THE PROJECT

This project focuses on developing an efficient and reliable lane detection system for Advanced Driver Assistance Systems (ADAS) using image-based inputs. The primary objective is to accurately identify lane boundaries under varying environmental conditions such as shadows, illumination changes, noise, and curved road structures while maintaining real-time performance. The system aims to improve detection stability by maintaining consistency across consecutive inputs, thereby minimizing fluctuations and instability in lane detection results.

The proposed work emphasizes the accurate detection of both straight and curved lane structures, which are commonly encountered in real-world driving scenarios. Another key objective is to reduce computational complexity while ensuring high detection accuracy, making the system suitable for resource-constrained environments. The design also focuses on compatibility with FPGA-based hardware implementation, enabling high-speed processing, low power consumption, and efficient real-time performance.

1.2 SCOPE OF THE PROJECT

This work focuses on vision-based lane detection using image processing techniques to enhance driver assistance functionality in modern vehicles. The system processes road images to extract and identify lane boundary features under different environmental conditions. The implementation is carried out using Python with OpenCV for software-level processing, enabling flexibility in development, testing, and validation of the algorithm.

In addition, FPGA implementation using VHDL is considered for hardware realization, providing high-speed execution and reduced power consumption. This dual implementation approach enables performance

evaluation in both software and hardware domains. The scope of the project is limited to lane detection and does not include full autonomous driving functionality, nor does it involve additional sensor integration such as LiDAR, radar, or GPS, focusing solely on image-based lane detection.

1.3 OUTLINE OF THE THESIS

This thesis is organized into multiple chapters, each addressing a specific aspect of the proposed lane detection system. The introduction chapter presents the background, motivation, objectives, and scope of the work, followed by the literature survey, which reviews existing techniques and highlights their limitations. The system overview chapter explains the architecture, data flow, and module organization, while the methodology chapter describes the complete processing pipeline, including preprocessing, ROI extraction, edge detection, feature extraction, and lane modeling.

The software implementation chapter details the Python-based development and validation of the system, whereas the hardware implementation chapter focuses on FPGA realization using VHDL for real-time performance. The experimental results and discussion chapter evaluates the system under different conditions using various performance metrics. Finally, the conclusion and future work chapter summarizes the contributions and suggests possible improvements for further enhancement.

1.4 OVERVIEW OF THE PROJECT

This project belongs to the domain of computer vision integrated with embedded systems, specifically focusing on applications in Advanced Driver Assistance Systems (ADAS). The system is designed to detect lane boundaries from road images using efficient image processing techniques. The proposed approach emphasizes reliability, stability, and real-time performance under

varying environmental conditions. It is designed to operate using only image-based inputs, making the system simple and cost-effective.

The system supports both software and hardware implementation, enabling execution using Python for development and FPGA for high-speed processing. This flexibility allows the system to be adapted for real-time embedded applications. Overall, the project aims to provide an efficient and scalable solution for lane detection, contributing to the advancement of intelligent transportation systems.

1.5 PROPOSED SOLUTION AND ADVANTAGES

The proposed system introduces an adaptive lane detection framework that enhances both accuracy and stability using image-based inputs. It utilizes Sliding Discrete Cosine Transform (SDCT) for effective edge detection, which provides better noise resistance and improved feature extraction even under low-contrast and varying illumination conditions. Blob analysis is used to group connected edge pixels into meaningful segments, facilitating accurate feature representation. Random Sample Consensus (RANSAC) is applied for robust lane modeling by eliminating outliers and fitting reliable lane curves, enabling accurate detection of both straight and curved lanes. A temporal consistency mechanism is incorporated to maintain continuity across consecutive inputs, reducing fluctuations and improving the stability of detection results.

The proposed approach offers several advantages, including high detection accuracy, improved robustness against noise and illumination variations, and enhanced stability across multiple inputs. It reduces computational requirements compared to deep learning-based methods, making it suitable for real-time applications. Furthermore, the system supports efficient implementation on embedded hardware such as FPGA, enabling faster processing, lower power consumption, and scalability for practical ADAS deployment.

CHAPTER 2

LITERATURE SURVEY

2.1 Title: Real-Time Lane Detection System Using Edge Detection and Hough Transform

Author Names: John McDonald, Peter Smith

Year: 2018

Description:

This paper presents a traditional lane detection approach based on spatial domain image processing techniques. The system mainly uses edge detection algorithms such as Canny and Sobel to extract lane boundary information from road images. After detecting edges, the Hough Transform is applied to identify straight lane lines by mapping image points into a parameter space and estimating line equations.

The method is simple, computationally efficient, and easy to implement, making it suitable for early driver assistance applications. However, it assumes that lane markings are clearly visible and mostly straight, which limits its effectiveness in real-world driving conditions. In situations such as curved roads, faded lane markings, shadows, and occlusions, the detection accuracy decreases significantly.

Moreover, the approach is sensitive to noise and lighting variations, which affects its performance in dynamic environments. It has limited capability in handling complex road geometries, especially when lane markings are unclear or discontinuous. Additionally, the absence of temporal consistency leads to unstable detection results across consecutive inputs. These limitations highlight the need for more adaptive and robust techniques suitable for real-world applications.

2.2 Title: Robust Lane Detection Using Sliding Discrete Cosine Transform (SDCT)

Author Names: Li Wei, Chen Zhang
Year: 2020

Description:

This research introduces a frequency-domain-based lane detection method using Sliding Discrete Cosine Transform (SDCT). Unlike conventional edge detection techniques, SDCT analyzes image data in the frequency domain, allowing effective extraction of both high-frequency and low-frequency components.

The method enhances lane visibility by detecting both prominent and subtle edges while reducing noise interference. This makes it highly effective in challenging conditions such as low illumination, shadow regions, and degraded lane markings. Compared to traditional operators like Sobel and Canny, SDCT provides improved edge detection performance. The computational complexity of SDCT is higher than spatial domain methods, and achieving real-time performance depends on proper optimization and hardware support. By preserving significant frequency components related to lane structures, SDCT improves feature representation and detection reliability in complex environments.

In addition, SDCT maintains consistency in feature extraction even when image quality varies significantly. This makes it suitable for applications requiring high robustness. Nevertheless, efficient implementation strategies are required to balance accuracy and processing speed in practical systems. This makes SDCT a reliable choice for advanced lane detection systems requiring high accuracy and robustness.

2.3 Title: Lane Detection and Tracking Using RANSAC-Based Model Fitting

Author Names: Ahmed Khan, Robert Lee

Year: 2019

Description:

This paper proposes a lane detection approach based on the Random Sample Consensus (RANSAC) algorithm for model fitting. The method estimates lane boundaries by repeatedly selecting subsets of feature points and fitting a mathematical model that best represents the lane structure. A key advantage of this technique is its robustness in handling noisy data and eliminating outliers. Even when the input contains incorrect or irrelevant feature points, RANSAC can still generate an accurate and stable model, making it suitable for real-world scenarios where lane markings may be partially missing or occluded.

The algorithm works by iteratively testing multiple candidate models and selecting the one that best fits the majority of the data points. This process helps filter noise and improves the reliability of the detected lane structure. RANSAC is particularly effective when the dataset contains a significant amount of outliers, which is common in road images due to shadows, vehicles, and road surface variations.

However, the performance of the algorithm depends on the quality of input features obtained from edge detection. Weak feature extraction can reduce model accuracy. Additionally, the iterative nature of RANSAC increases computational cost, which may impact real-time performance. The method also lacks temporal continuity, which can result in variation across consecutive inputs. Furthermore, uneven distribution of feature points may affect model estimation, and the absence of adaptive tuning limits flexibility under varying conditions. Integrating RANSAC with improved feature extraction methods can enhance overall system performance.

2.4 Title: Temporal Consistency in Lane Detection for Autonomous Driving

Author Names: David Brown, Kim Lee

Year: 2022

Description:

This research focuses on improving the stability of lane detection systems by introducing temporal consistency across consecutive frames. Instead of treating each image independently, the method considers previous frame information to smooth lane predictions over time. This approach significantly reduces flickering and improves the continuity of lane detection in real-time driving environments. It enhances system reliability, especially in scenarios involving rapid motion or sudden environmental changes. The method ensures that detected lane lines remain stable across multiple frames, improving visual consistency and overall system usability.

Implementing temporal consistency introduces additional computational overhead and requires memory management to store previous frame data. Despite this limitation, the method greatly improves the robustness of lane detection systems in dynamic conditions and is widely used in modern ADAS frameworks. Proper parameter tuning is also required to balance stability and responsiveness. Incorrect tuning may result in delayed adaptation to sudden lane changes. In some cases, excessive smoothing may reduce the system's ability to detect rapid transitions. Therefore, careful optimization is essential for achieving reliable performance.

Maintaining historical data for multiple frames increases memory requirements in embedded systems. The method must also ensure synchronization between frames to avoid inconsistencies. Efficient buffering techniques are necessary to handle continuous data streams. These considerations are important for achieving real-time implementation.

2.5 Title: Lightweight Lane Detection System for Embedded FPGA Implementation

Author Names: Sunil Kumar, James Wilson

Year: 2023

Description:

This paper presents a lightweight lane detection framework optimized for FPGA-based embedded systems. The focus of the study is to reduce computational complexity while maintaining acceptable accuracy for real-time applications. The system divides lane detection tasks into parallel hardware modules, enabling high-speed processing. Techniques such as simplified edge detection and optimized lane modeling are used to ensure compatibility with hardware constraints. The FPGA implementation allows concurrent execution of operations, significantly improving processing speed and reducing latency.

While the system achieves fast processing speed and low power consumption, it has limitations in handling highly complex road environments such as sharp curves, heavy occlusions, and poor lighting conditions. However, it is highly suitable for automotive embedded systems where real-time performance and energy efficiency are critical design requirements. Additionally, the hardware-oriented design ensures efficient utilization of FPGA resources while maintaining system scalability.

Furthermore, optimization techniques such as pipelining and parallel processing improve efficiency. The system must consider hardware constraints such as memory bandwidth and logic utilization for efficient implementation. Overall, the approach supports efficient real-time implementation in resource-constrained embedded environments.

CHAPTER 3

EXISTING SYSTEM

3.1 OVERVIEW OF EXISTING SYSTEM

The existing lane detection systems were mainly developed using traditional image processing and classical computer vision techniques to identify lane boundaries from road images. These systems were designed to support basic driver assistance functions in Advanced Driver Assistance Systems (ADAS) by detecting lane markings and providing simple guidance for navigation. The general processing pipeline consisted of image preprocessing, edge detection, and line fitting to extract lane structures from input road images. In conventional approaches, the input image was first converted into grayscale format to reduce computational complexity and simplify further processing.

Noise reduction techniques such as Gaussian filtering were commonly applied to enhance image quality and suppress unwanted variations. After preprocessing, edge detection was performed using operators such as Sobel, Prewitt, and Canny. These operators detected significant intensity changes in the image, which typically corresponded to lane markings. Among these, the Canny edge detector was widely used due to its better edge localization and improved noise suppression capability.

After edge detection, the resulting edge map was processed using the Hough Transform technique to detect straight lines representing lane boundaries. The Hough Transform converted edge points from image space into a parameter space, where aligned points accumulated votes to identify dominant lines. The lines with the highest votes were selected as lane boundaries. In improved versions, Probabilistic Hough Transform (PHT) was used to reduce computational complexity and improve efficiency. Additionally, Region of

Interest (ROI) selection was applied to focus only on road regions and remove irrelevant background elements such as vehicles, trees, and roadside objects.

Some enhanced traditional systems also incorporated simple rule-based logic or basic machine learning techniques to improve detection performance. However, the majority of existing systems still relied on deterministic geometric and mathematical approaches. These methods were preferred in embedded applications due to their low computational cost, ease of implementation, and real-time capability under controlled conditions. However, their performance was highly dependent on environmental conditions and predefined assumptions regarding lane structure.

3.2 LIMITATIONS OF EXISTING SYSTEM

Existing lane detection systems suffer from limitations due to reliance on fixed algorithms and sensitivity to environmental conditions, reducing their effectiveness in real-world scenarios. These challenges highlight the need for more adaptable and reliable approaches to improve overall performance.

3.2.1 Sensitivity to Illumination Changes

Existing systems are highly sensitive to variations in lighting conditions such as shadows, sunlight glare, nighttime environments, and tunnels. These changes significantly affect image quality and reduce the accuracy of edge detection algorithms. As a result, lane markings may not be detected properly, leading to incorrect or missing lane boundaries.

3.2.2 Noise Sensitivity

Real-world road images often contain noise caused by camera limitations, motion blur, vibrations, and environmental factors such as rain, fog, and dust. This noise introduces unwanted edges and distortions in the image, which leads to false detection of lane markings and reduces overall system reliability.

3.2.3 Lack of Temporal Consistency

Most existing lane detection systems process each frame independently without considering information from previous frames. This results in unstable detection outputs, causing flickering or sudden changes in detected lane positions. Such instability reduces driver confidence and system reliability.

3.2.4 Occlusion Problems

Lane detection is often affected by occlusions caused by vehicles, pedestrians, trees, shadows, and roadside objects. These obstacles can partially or completely block lane markings, leading to incomplete or fragmented detection results.

3.2.5 Moderate Accuracy in Real-World Conditions

Although existing systems perform well in controlled environments, their accuracy decreases significantly in real-world conditions. Typically, these systems achieve accuracy between 70 percent and 85 percent, which is not sufficient for safety-critical applications like ADAS.

3.2.6 Limited Adaptability to Environmental Variations

Existing systems are designed based on fixed thresholds and assumptions, making them less adaptable to varying road conditions, weather changes, and lighting variations. This lack of adaptability leads to inconsistent performance across different environments. This limitation reduces the system’s ability to perform reliably in diverse real-world driving scenarios.

3.2.7 High Computational Complexity in Advanced Methods

Deep learning-based lane detection methods improve accuracy but require large datasets, high computational power, and specialized hardware such as GPUs. This makes them unsuitable for embedded systems and FPGA-based real-time

implementations where resources are limited. Additionally, these approaches increase system cost and power consumption, making them less practical for low-power automotive applications.

3.2.8 Poor Handling of Complex Road Scenarios

Existing systems struggle in complex situations such as intersections, lane merging, lane splitting, and multi-lane highways. The presence of multiple overlapping lane markings makes it difficult for traditional algorithms to correctly identify lane boundaries.

3.2.9 Dependency on Clear Lane Markings

Most traditional lane detection methods rely heavily on clearly visible lane markings. However, in real-world scenarios, lane markings may be faded, worn out, or partially visible. This significantly reduces detection accuracy and system performance.

3.2.10 Limited Robustness to Camera Variations

Different cameras may produce images with varying resolutions, angles, and distortions. Existing systems are not always robust to these variations, which affects their ability to generalize across different setups.

3.2.11 Insufficient Feature Extraction Techniques

Traditional edge detection methods capture only basic features and may fail to extract complex patterns from the image. This limits the system's ability to accurately detect lane boundaries under challenging conditions. As a result, the system may fail to detect lanes accurately in complex scenarios. Advanced feature extraction techniques are required to improve detection performance.

CHATPER 4

PROPOSED SYSTEM

4.1 OVERVIEW OF PROPOSED SYSTEM

The proposed system introduces an adaptive and temporally consistent lane detection framework designed to overcome the limitations of existing techniques and improve performance in Advanced Driver Assistance Systems (ADAS). It focuses on achieving high accuracy, robustness, temporal stability, and computational efficiency using image-based inputs. Unlike traditional spatial domain methods and heavy deep learning models, the proposed approach combines frequency-domain processing, statistical modeling, and temporal consistency for reliable lane detection in real-world conditions.

The system begins with preprocessing, where the input road image is converted into grayscale to reduce computational complexity and retain essential structural information. A Region of Interest (ROI) is then applied to focus only on the road area, eliminating irrelevant background regions such as sky, trees, and buildings, thereby improving efficiency and accuracy. Edge detection is performed using the Sliding Discrete Cosine Transform (SDCT), which operates in the frequency domain and effectively detects both strong and weak lane edges while reducing noise and handling illumination variations. After edge extraction, blob analysis is applied to group connected pixels into meaningful lane segments and remove isolated noise points.

For lane modeling, the Random Sample Consensus (RANSAC) algorithm is used to fit robust lane boundaries by eliminating outliers and selecting the best-fitting model from multiple iterations. A key enhancement of the proposed system is temporal consistency, which ensures stable lane detection across consecutive frames by reducing flickering and sudden variations. Additionally, adaptive lane modeling enables accurate detection of both straight and curved lanes, making the system suitable for complex road environments.

4.2 ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers several advantages over existing methods. It provides improved accuracy by combining SDCT-based edge detection with RANSAC-based lane modeling, ensuring reliable performance even under noise, shadows, and illumination changes. Temporal consistency is another major advantage, as it stabilizes lane detection across frames and eliminates flickering issues, resulting in smooth and continuous lane tracking essential for real-time ADAS applications.

The system is also highly adaptable, capable of detecting both straight and curved lanes effectively, making it suitable for highways, urban roads, and complex road geometries. It is computationally efficient as it avoids deep learning complexity, large datasets, and GPU requirements, making it ideal for embedded systems and FPGA-based implementations. This significantly reduces processing overhead and enables faster execution in real-time environments. The system maintains reliable performance under varying environmental conditions such as low light, fog, and noisy road images SDCT enhances edge quality while RANSAC effectively removes outliers, improving reliability. The architecture is optimized for real-time performance, ensuring fast processing suitable for safety-critical applications. Furthermore, it supports both software (Python/OpenCV) and hardware (FPGA/VHDL) implementation, making it flexible and scalable for automotive systems.

Another important advantage is its low memory requirement compared to deep learning approaches, which makes it suitable for resource-constrained devices. The modular design of the system allows easy modification and integration with other ADAS functionalities. Additionally, the system ensures consistent performance across different road scenarios without requiring extensive training data. This makes the proposed approach both cost-effective and practical for real-world deployment.

4.3 APPLICATIONS OF PROPOSED SYSTEM

The proposed lane detection system is widely used in modern automotive applications, especially in Advanced Driver Assistance Systems (ADAS), where it assists drivers in maintaining proper lane positioning and improves overall road safety. It also plays an important role in autonomous driving systems by providing accurate lane information required for vehicle navigation and control. The system is highly useful in highway driving conditions, where it helps maintain lane alignment during long-distance travel and reduces driver fatigue by providing consistent lane guidance.

In addition to automotive applications, the system can be integrated into Intelligent Transportation Systems (ITS) for traffic monitoring, road condition analysis, and infrastructure planning. It enables efficient extraction of lane-related data, which can be used for traffic flow analysis and road maintenance. The system can also be applied in urban driving environments, where complex road geometries and multiple lanes require reliable detection. Furthermore, it can be used in surveillance systems to monitor lane discipline and detect traffic violations, contributing to better traffic management.

Due to its low computational complexity, the proposed system is suitable for embedded platforms and FPGA-based hardware implementations used in modern vehicles. It can also be utilized in driver safety systems to monitor lane positioning and provide real-time assistance. Additionally, the system can be extended to smart city applications for automated traffic analysis and efficient road usage monitoring. It is also useful in driver training simulators and research environments for testing and evaluating lane detection algorithms, making it a versatile solution for future intelligent transportation systems.

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB
Storage	256 GB SSD or higher
Input Source	Road Images
FPGA Design Tool	Quartus Prime lite
Hardware Description Language	VHDL
Display Device / Platform	Laptop Screen / FPGA Remote lab

Table 5.1 Hardware requirements of the proposed system

As specified in Table 5.1 Hardware Requirements of the Proposed System, the lane detection framework requires a processor with adequate computational capability to support image processing tasks such as preprocessing, edge detection, feature extraction, and lane modeling. A minimum of 8 GB RAM is required for smooth execution of image processing operations, along with a 256 GB SSD or higher for storing datasets, intermediate results, and output images. For hardware implementation, an FPGA platform programmed using VHDL and Quartus Prime Lite is used. The FPGA enables parallel processing, improving execution speed and making the system suitable for real-time applications. The detected lane output is displayed on a laptop during software execution and verified through FPGA remote laboratory setups for real-time validation.

5.2 SOFTWARE REQUIREMENTS

Software Component	Technology Used
Programming Language	Python
Image Processing Library	OpenCV
Numerical Computation	NumPy
Visualization Tool	Matplotlib
Operating System	Windows
Development Tool	Visual Studio Code

Table 5.2 Software requirements of the proposed system

As presented in Table 5.2 Software Requirements of the Proposed System, the lane detection system is developed using a hybrid software and hardware design environment to support efficient image processing and FPGA implementation. The software module is implemented using Python as the primary programming language due to its simplicity and strong support for computer vision applications.

Image processing tasks such as grayscale conversion, region of interest extraction, and edge detection are performed using the OpenCV library. Numerical computations required for transformations and lane modeling are handled using NumPy, while Matplotlib is used for visualizing the detected lane outputs. For hardware implementation, VHDL is used to design the FPGA logic, and Quartus Prime Lite is used for synthesis, simulation, and deployment on FPGA hardware. The system is executed on Windows or Linux operating systems, and development is carried out using tools such as Visual Studio Code or Jupyter Notebook.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 SYSTEM ARCHITECTURE

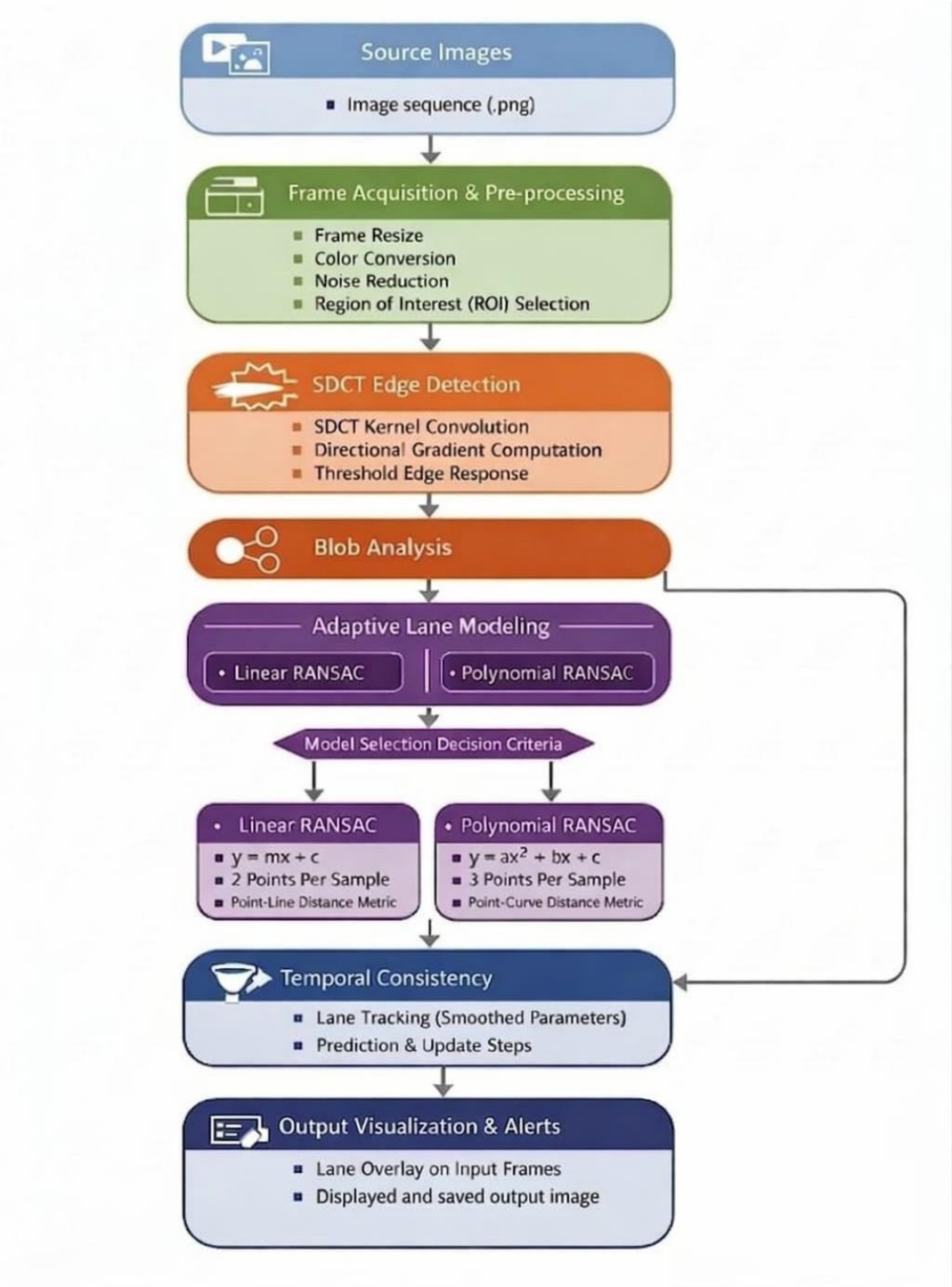

Fig 6.1 Architecture design of lane detection framework

The architectural design of the Lane Detection Framework illustrates the structural organization and interaction among the major components involved in the image-based lane detection system. The architecture follows a modular pipeline design that integrates image acquisition, preprocessing, edge detection, feature extraction, lane modeling, temporal consistency, and output visualization components.

The architecture presented in Fig 6.1 illustrates the data flow between input images, processing modules, and output visualization. The interaction among these components ensures efficient processing of road images, accurate extraction of lane features, and reliable detection of lane boundaries under varying environmental conditions. The modular design also enables easy integration with real-time embedded systems and future enhancements.

6.1.1 Image Input Module

The Image Input Module acts as the initial stage of the system, where road images are provided as input for processing. These images can be obtained either from publicly available datasets or captured in real time using camera systems mounted on vehicles. The module is responsible for loading, resizing, and formatting the input images into a standard structure suitable for further processing.

Each input image represents a frame of the road scene containing lane markings, vehicles, and surrounding environmental elements. Proper handling of image resolution, format, and clarity is essential, as these factors directly affect the performance of subsequent modules. High-quality input images improve detection accuracy, while poor-quality images may lead to incorrect or incomplete lane detection. The module may also perform initial validation checks to ensure that the input data meets required specifications such as resolution and format consistency before forwarding it to the preprocessing stage.

6.1.2 Preprocessing Module

The preprocessing module prepares the input image for efficient analysis by reducing noise and simplifying the data. In this stage, the input image is first converted into grayscale format, which reduces computational complexity by eliminating unnecessary color information while preserving structural details. Noise reduction techniques such as filtering may be applied to remove unwanted variations caused by environmental conditions like lighting changes or camera noise.

Additionally, a Region of Interest (ROI) is selected to focus only on the relevant road portion of the image. By removing irrelevant regions such as the sky, buildings, and roadside objects, the system reduces processing time and minimizes false detections. This module plays a crucial role in enhancing the quality and reliability of the input data before further processing. Proper preprocessing ensures that the subsequent modules operate on clean and meaningful data, which significantly improves the overall efficiency and accuracy of the system.

6.1.3 Edge Detection Module (SDCT)

The edge detection module is responsible for extracting lane boundaries from the preprocessed image using the Sliding Discrete Cosine Transform (SDCT). Unlike conventional edge detection techniques that operate in the spatial domain, SDCT works in the frequency domain, enabling better identification of both strong and weak edges.

This approach improves the detection of lane markings even under challenging conditions such as shadows, low contrast, and uneven illumination. The sliding window mechanism of SDCT allows continuous analysis of image regions, improving edge localization and consistency. As a result, this module produces a refined edge map that clearly highlights lane structures while

effectively suppressing noise and irrelevant details. SDCT improves robustness by preserving important frequency components associated with lane features, which helps maintain detection accuracy even when image quality varies.

6.1.4 Feature Extraction Module (Blob Analysis)

The feature extraction module processes the edge-detected image to identify meaningful lane-related features. In this stage, blob analysis is applied to group connected edge pixels into clusters known as blobs. Each blob represents a potential lane segment or relevant structure in the image.

This method helps in filtering out isolated or insignificant edges that do not contribute to lane detection. By focusing only on prominent clusters, the system reduces false positives and improves detection accuracy. The extracted features serve as input for the lane modeling stage, ensuring that only relevant and structured data is used for further analysis. The use of blob-based grouping also simplifies the complexity of the dataset by reducing the number of feature points that need to be processed in later stages.

6.1.5 Lane Modeling Module (RANSAC)

The lane modeling module applies the Random Sample Consensus (RANSAC) algorithm to determine the best-fit lane boundaries from the extracted features. RANSAC works by iteratively selecting random subsets of data points and fitting a model to them, while eliminating outliers that do not conform to the expected pattern. This makes the method highly robust against noise, missing data, and irregular lane markings. The algorithm identifies the most consistent line or curve that represents the lane structure. It is capable of detecting both straight and curved lanes, making it suitable for real-world road conditions.This module ensures that the final lane model is accurate and reliable even in the presence of disturbances. Additionally, RANSAC improves the stability of lane

estimation by focusing only on inlier data points, thereby enhancing overall detection performance.

6.1.6 Temporal Consistency Module

The temporal consistency module improves the stability of lane detection across consecutive frames. Instead of processing each image independently, this module compares the current frame with previously detected lane data. By maintaining continuity in lane positions over time, the system reduces sudden changes, flickering, and instability in the output.

This is particularly important for real-time applications, where smooth and consistent lane detection is required for safe driving assistance. The module may use buffering or tracking techniques to store previous results and refine current predictions. As a result, it enhances the overall robustness and reliability of the system. It also helps in handling temporary detection failures by utilizing historical data to maintain continuity.

6.1.7 Output Visualization Module

The output visualization module generates the final result by overlaying the detected lane boundaries on the original input image. This provides a clear and intuitive representation of the detected lanes, making it easier for users or systems to interpret the results.

The module can display the output in real time for live applications or store the processed images and videos for further analysis and evaluation. Additional visual elements such as colored lane lines, markers, or confidence indicators may be used to improve clarity. This module plays an important role in presenting the system output in a user-friendly and understandable format. It also aids in performance evaluation by enabling visual verification of detection accuracy.

6.2 ALGORITHM USED IN THE PROPOSED SYSTEM

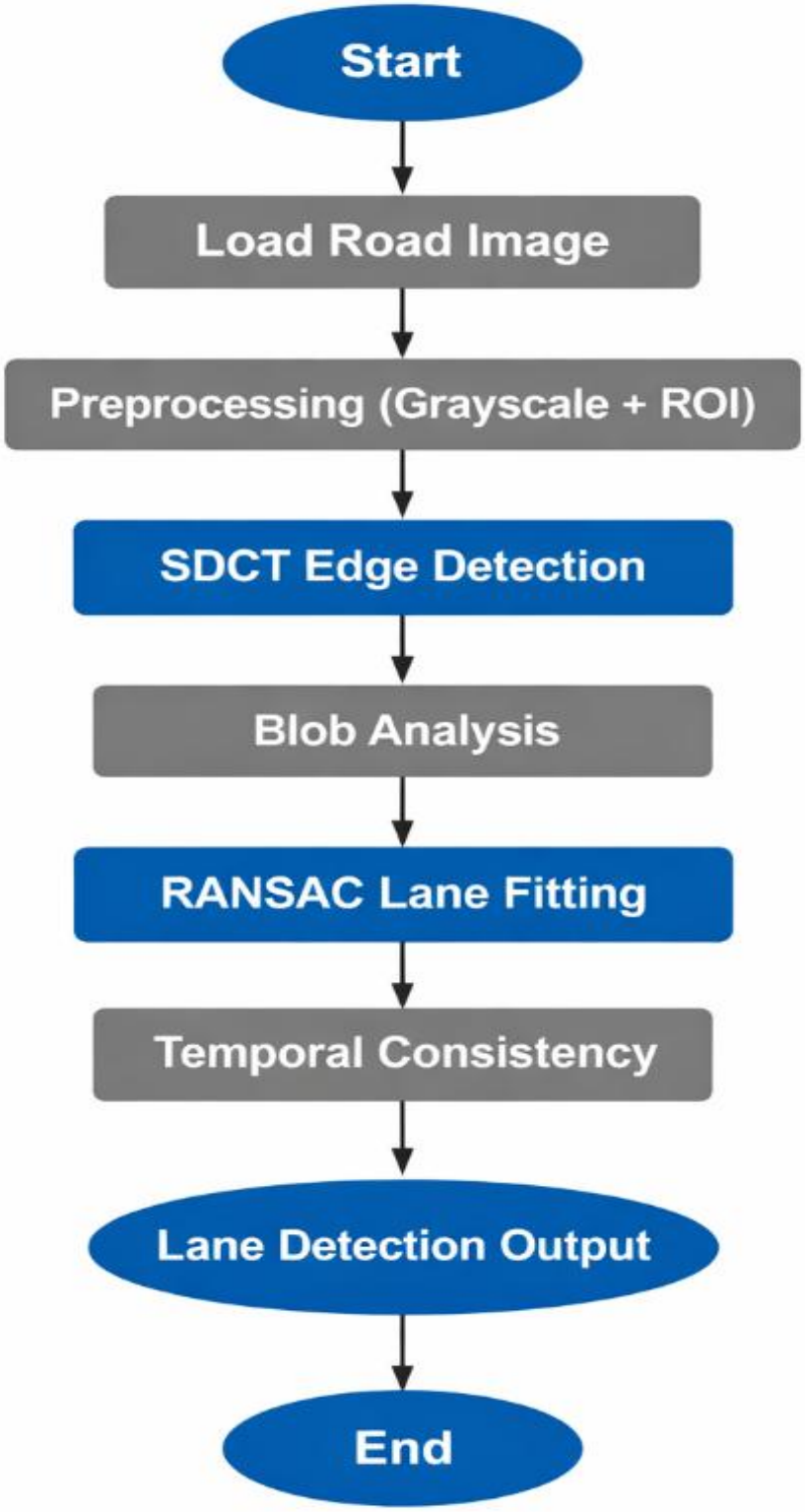

Fig 6.2 Algorithm of lane detection framework

Begin

1. Start the system
2. Load input road image from dataset or camera source
3. Convert RGB image into grayscale format
4. Apply Region of Interest (ROI) selection to extract road region
5. Perform noise reduction and image normalization (if required)
6. Apply Sliding Discrete Cosine Transform (SDCT) for edge detection
7. Extract lane boundary edges from SDCT output
8. Perform blob analysis to group connected edge pixels
9. Filter out small and irrelevant blobs (noise removal)
10. Apply Random Sample Consensus (RANSAC) algorithm
    - Randomly select sample points
    - Fit lane model (straight/curved)
    - Identify inliers and outliers
    - Optimize best lane fit
11. Validate lane model consistency
12. Apply temporal consistency between consecutive images
    - Compare current lane position with previous result
    - Apply smoothing filter to reduce variation
13. Generate final lane boundary estimation
14. Overlay detected lane on original image
15. Display output image

End

This algorithm ensures accurate, stable, and real-time lane detection under varying environmental conditions.

S.No	Algorithm	Description	Purpose in System
1	Sliding Discrete Cosine Transform (SDCT)	Frequency-domain edge detection method	Used to extract lane edges from road images even under noise and low illumination
2	RANSAC Algorithm	Robust statistical model fitting technique	Used to accurately fit lane boundaries and remove outliers
3	Blob Analysis	Image grouping technique	Used to extract meaningful lane segments from edge map
4	Temporal Consistency	Frame-to-frame smoothing technique	Used to maintain stable lane detection across consecutive images
5	Region of Interest (ROI)	Image focusing technique	Used to isolate road region and remove unwanted background
6	Grayscale Conversion	Preprocessing technique	Used to simplify image and reduce computational load

Table 6.1 Algorithms and Techniques Used in the Proposed System

6.3 FPGA DESIGN IMPLEMENTATION USING INTEL QUARTUS PRIME LITE

6.3.1 Overview of Quartus Environment

The FPGA design for the proposed lane detection system is developed using Intel Quartus Prime Lite. It is a comprehensive design tool that supports design entry, synthesis, simulation, and hardware programming for FPGA devices. The software provides an integrated development environment where users can create, compile, and analyze hardware designs efficiently.

The Quartus environment consists of multiple modules including project management, design compilation, waveform simulation, and device programming. It enables designers to convert high-level hardware descriptions written in VHDL into optimized hardware configurations suitable for FPGA implementation. This ensures accurate mapping of the design onto FPGA resources. Additionally, the tool provides detailed reports and analysis features that help in evaluating resource utilization and timing performance of the design.

6.3.2 Project Creation and Design Entry

The design process begins with the creation of a new project in Intel Quartus Prime Lite. The project setup includes specifying the project name, working directory, and target FPGA device. The appropriate device is selected based on the hardware platform used for implementation. Proper configuration of project settings ensures compatibility between the design and the target FPGA architecture.

Once the project is created, the VHDL design files corresponding to the lane detection system are added. These files define the functional behaviour of different modules such as preprocessing, edge detection, and lane extraction. The design entry stage ensures that all modules are properly integrated before

compilation. This stage also facilitates hierarchical design organization, improving readability and maintainability of the overall system. It further enables modular development, allowing individual components to be tested and updated independently.

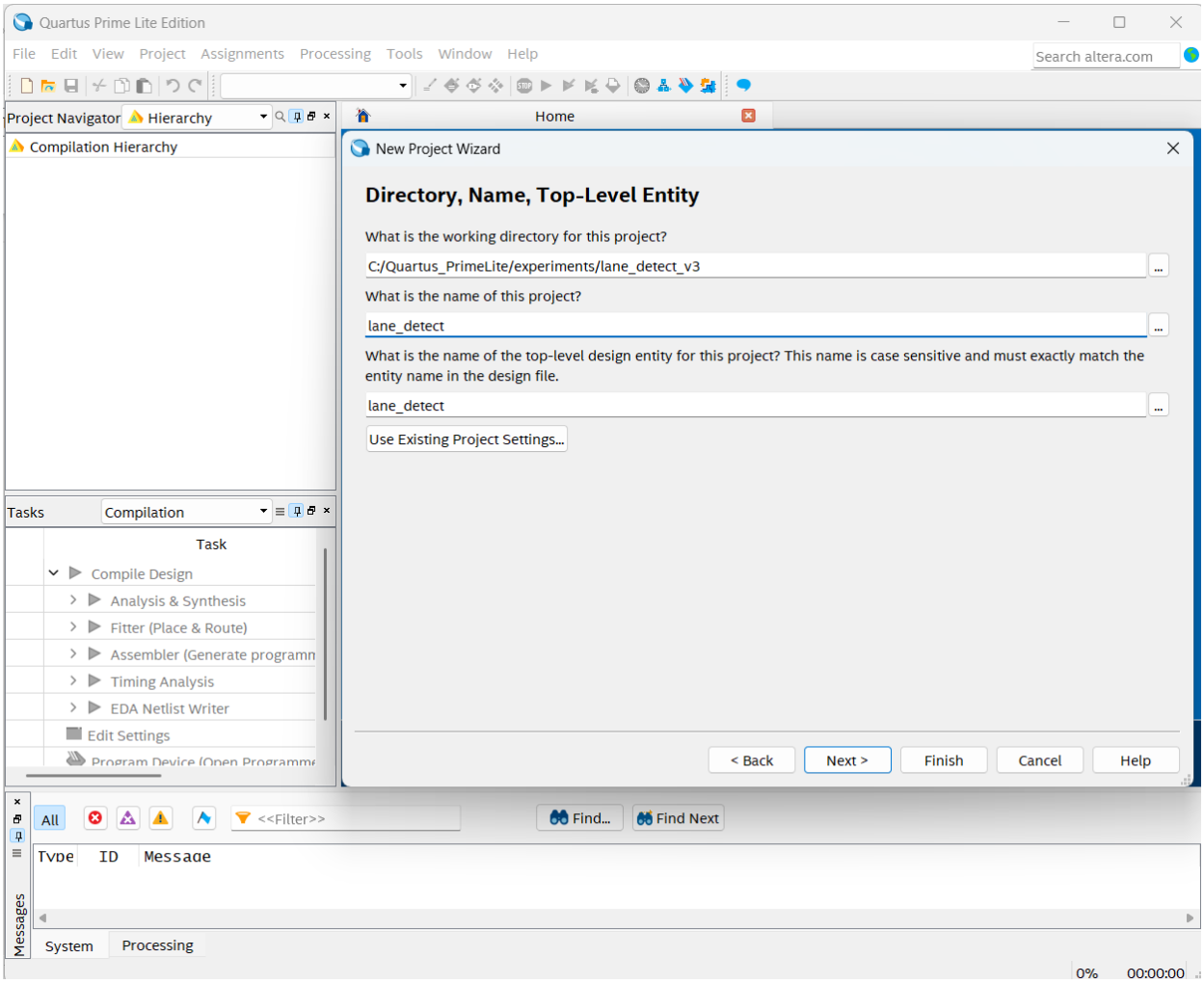

Fig 6.3 Quartus Project Creation and Design Entry

6.3.3 Design Compilation and Synthesis

Following design entry, the project is compiled using the Intel Quartus Prime Lite compiler. The compilation process includes synthesis, placement, routing, and timing analysis. During synthesis, the VHDL code is translated into a gate-level representation, which is then optimized for FPGA implementation. This stage ensures that the logical functionality of the design is accurately

transformed into hardware-level structures. Advanced optimization techniques are applied to reduce logic usage and improve performance.

It maps the design onto logic elements, registers, and interconnects within the FPGA. Any errors or warnings are identified during this stage and must be resolved before proceeding further. Successful compilation indicates that the design is ready for hardware execution. The timing analysis step verifies that all signal paths meet required timing constraints, ensuring reliable system operation. This overall process guarantees that the design is both functionally correct and optimized for efficient hardware utilization. It also helps in identifying critical paths that may affect system speed.

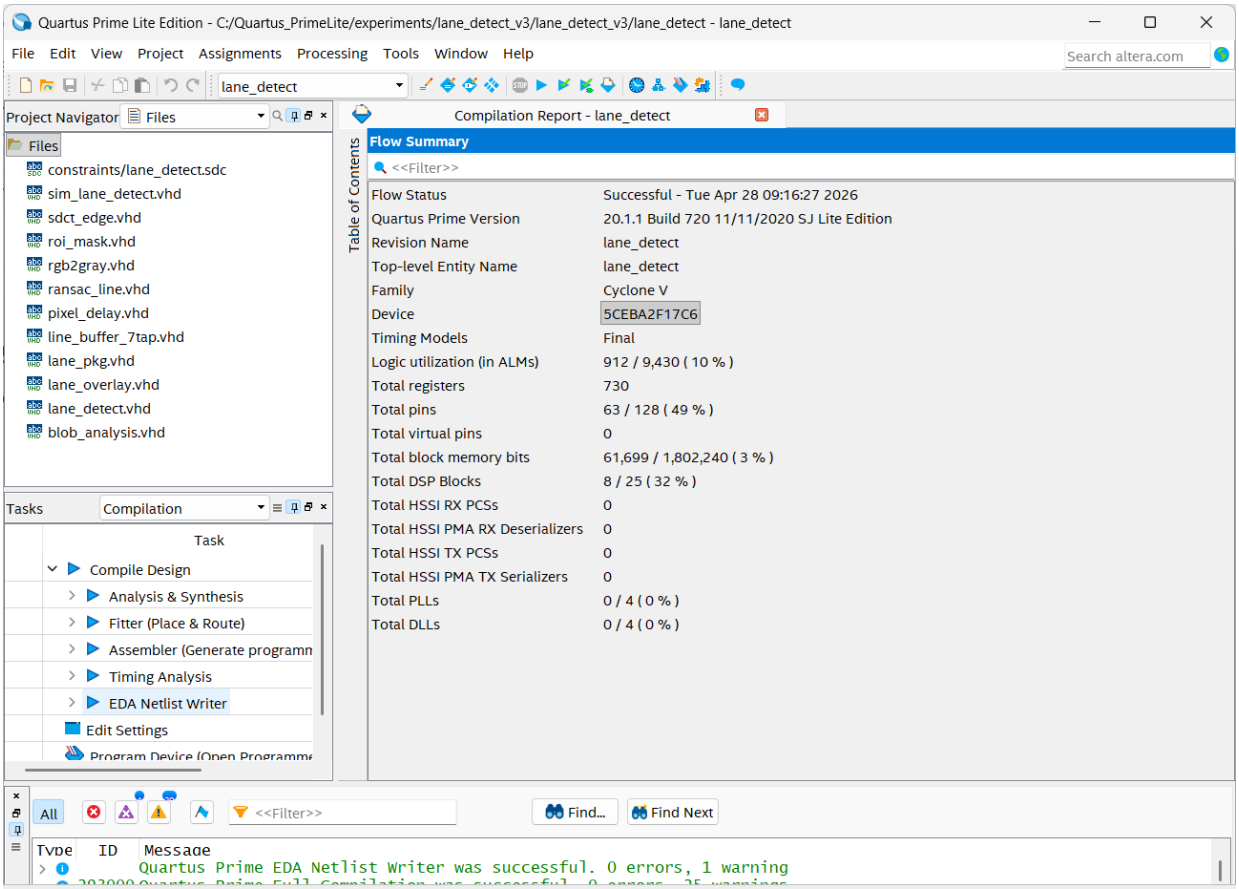

Fig 6.4 Quartus Compilation Process

6.3.4 Simulation and Functional Verification

Simulation is performed to verify the correctness of the design before hardware implementation. Intel Quartus Prime Lite provides waveform simulation tools that allow observation of signal transitions and timing behaviour. This enables detailed analysis of how input signals propagate through different stages of the design.

Test inputs are applied to the design, and the output waveforms are analyzed to ensure proper functionality of each module. This step ensures logical correctness before hardware implementation and helps detect any design issues prior to FPGA deployment. Simulation also allows verification of edge cases and timing constraints, ensuring reliable operation under different conditions. This process reduces the risk of errors during actual hardware execution.

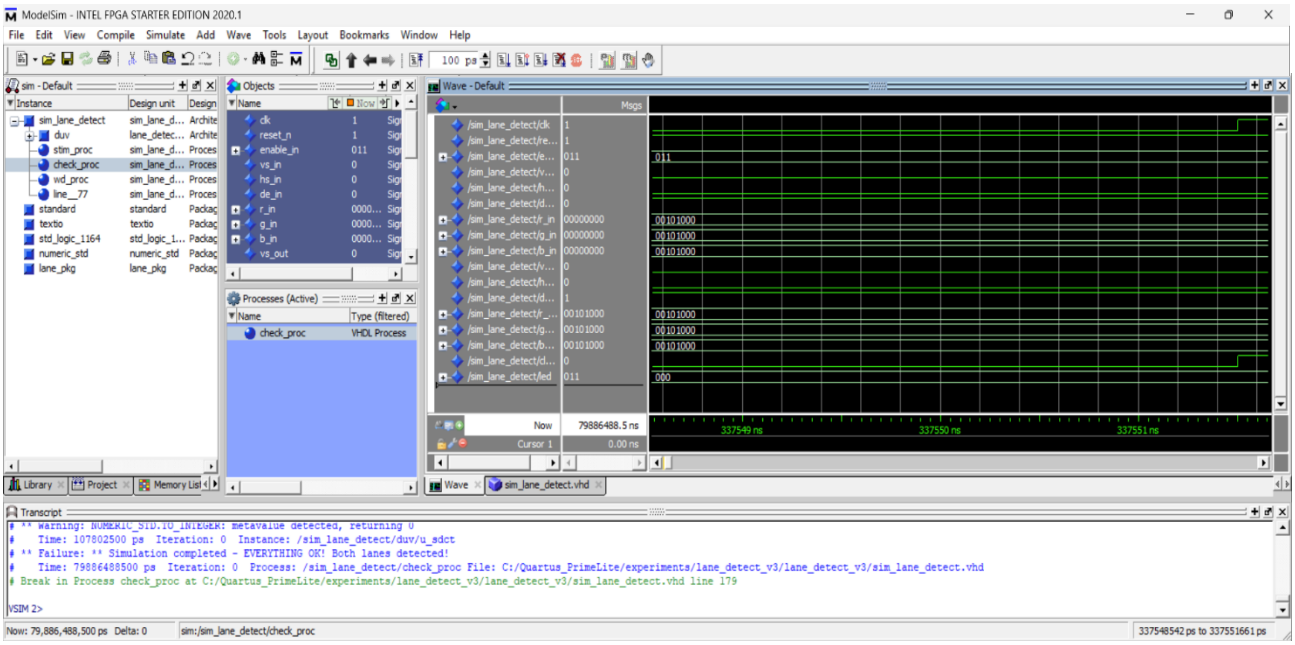

Fig 6.5 Simulation Waveform Output

6.3.5 FPGA Programming File Generation

Once the design is successfully compiled and verified, the programming file is generated using Intel Quartus Prime Lite. The tool produces a configuration file in the form of a .sof (SRAM Object File), which contains the synthesized

hardware implementation of the lane detection system. This file represents the final mapping of the design onto FPGA resources.

The generated .sof file is uploaded to the remote FPGA platform for execution, eliminating the need for direct hardware programming from the local system. It encapsulates all logic, routing, and configuration data required for FPGA operation. This approach simplifies deployment while ensuring accurate and efficient hardware execution of the design. The use of remote programming also enhances accessibility and allows testing without requiring physical hardware setup.

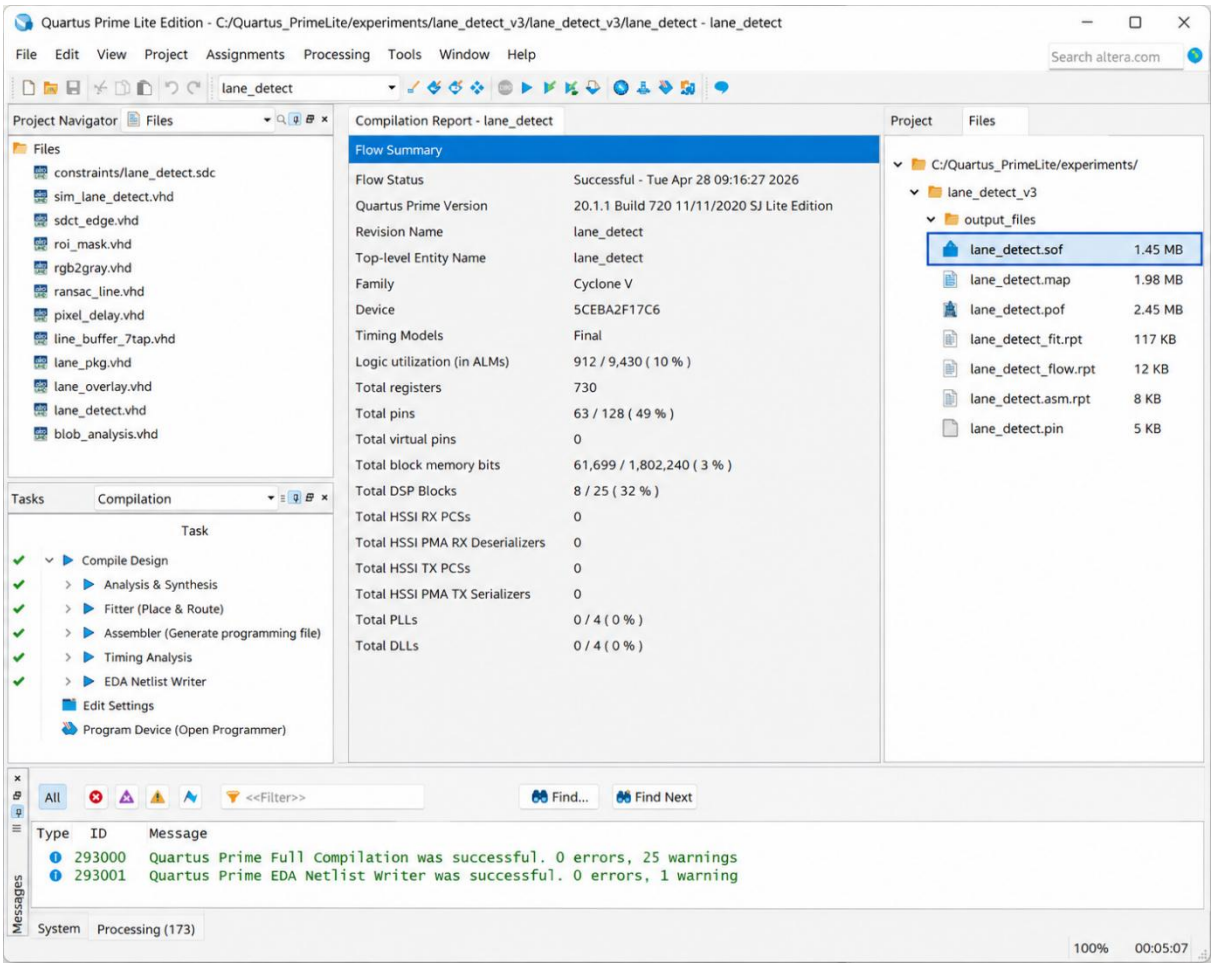

Fig 6.6 FPGA Programming File Generation

6.4 FPGA REMOTE LAB IMPLEMENTATION

6.4.1 Overview of Remote Lab Environment

The FPGA Vision Remote Lab is a web-enabled hardware experimentation platform that allows users to remotely access and configure real FPGA devices through an internet interface. The system is built on a centralized server architecture where FPGA development boards are physically installed and connected to backend control units. These backend systems manage user requests, handle communication, and coordinate hardware-level operations, thereby enabling seamless interaction between remote users and the FPGA hardware.

Through this platform, users can upload synthesized hardware designs, configure experimental parameters, and execute them in real time without direct physical access to the device. The overall system consists of a browser-based interface, a server-side management layer, and the FPGA hardware unit. Input data is transmitted to the FPGA, processed using a parallel and pipelined architecture, and the results are returned to the user interface with minimal latency. This setup ensures efficient resource utilization and supports real-time applications such as image processing and embedded system validation.

6.4.2 Remote Lab Access Procedure

Access to the remote FPGA platform is provided through a secure web-based interface that ensures controlled usage of shared hardware resources. The system requires user authentication, where valid credentials are entered through a browser-based portal. Upon successful login, the user is redirected to a centralized dashboard that provides access to FPGA configurations, experiment controls, and system status information.

From a system-level perspective, the web interface functions as a communication layer between the user and the FPGA hardware. It manages the transfer of configuration files, input data, and output results through the server. This abstraction simplifies hardware interaction and allows users to perform complex FPGA-based experiments remotely without direct physical access.

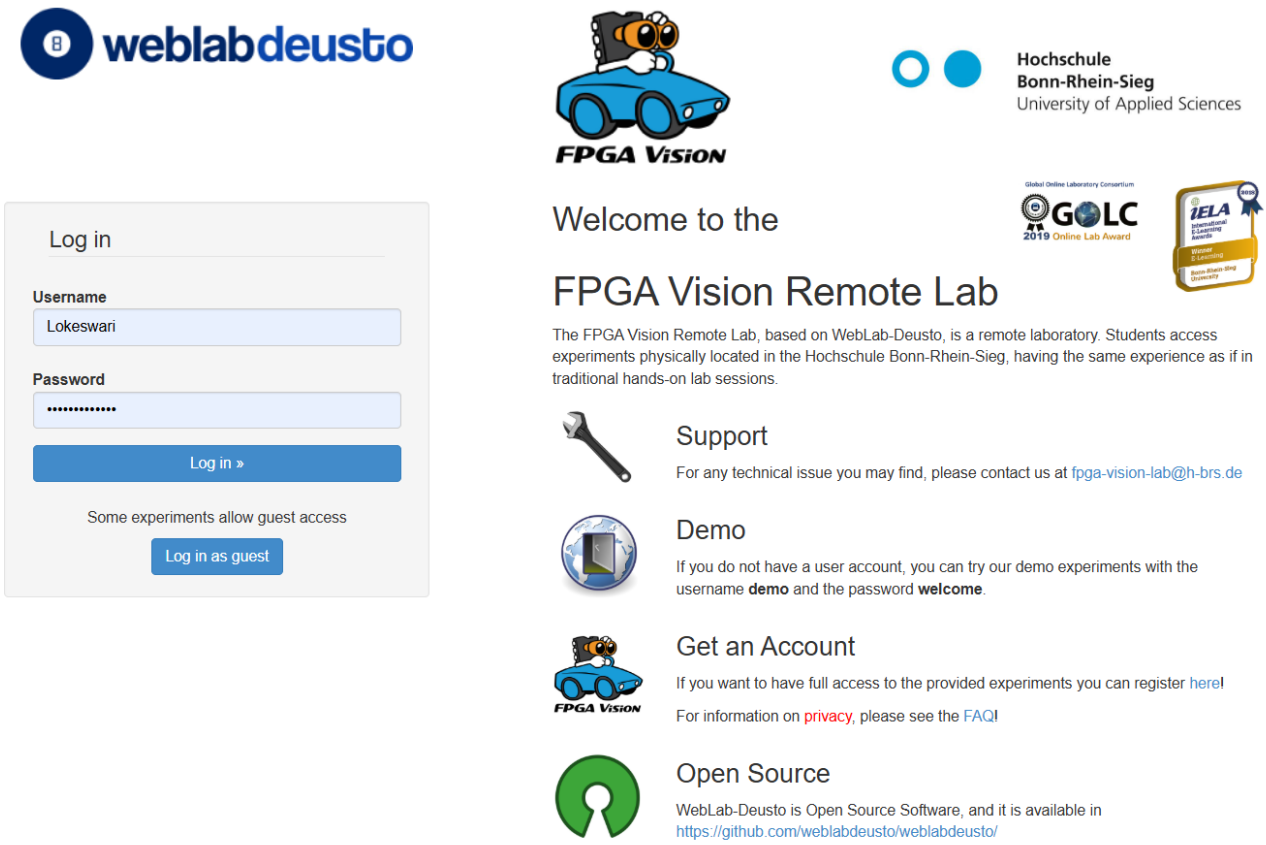

Fig 6.7 FPGA Remote lab login interface

6.4.3 Experiment Selection and Reservation

After authentication, the user selects an appropriate FPGA configuration based on the application requirements. The platform provides multiple configurations such as CV1 and CV2, which correspond to different hardware setups and capabilities. Once selected, the FPGA resource is reserved through a scheduling mechanism that allocates exclusive access for a defined duration.

This reservation system plays a crucial role in multi-user environments by ensuring efficient resource utilization and preventing conflicts. Upon successful

reservation, the FPGA is initialized and prepared to accept the hardware configuration. This guarantees stable execution and consistent system performance throughout the experiment.

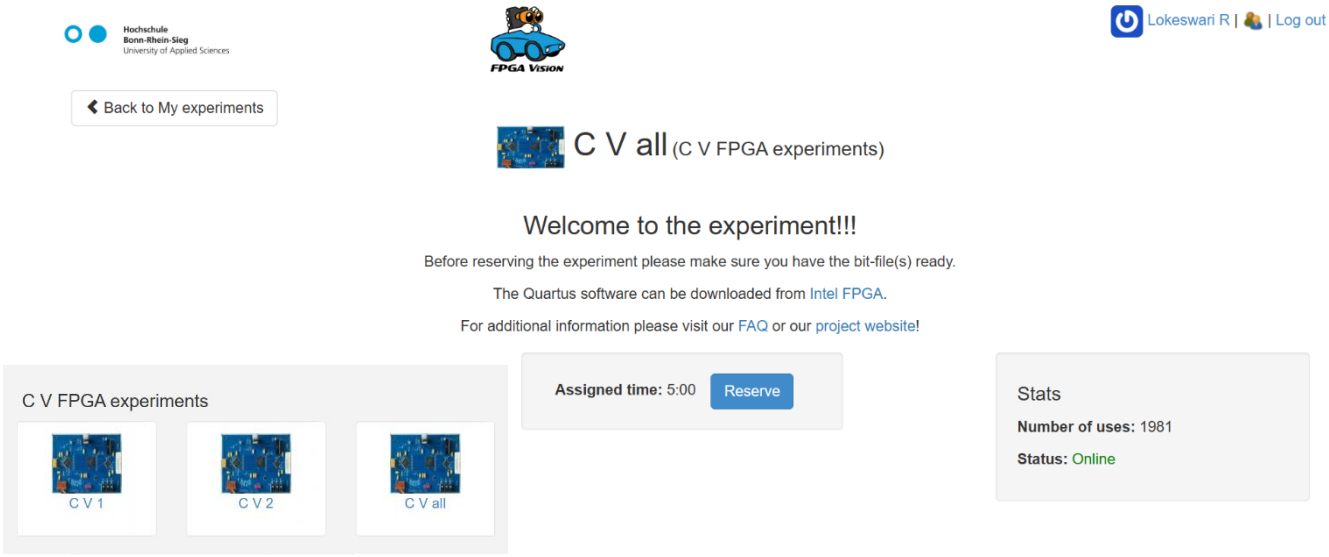

Fig 6.8 FPGA resource reservation interface

6.4.4 FPGA Binary Upload and Execution

The FPGA implementation is carried out by uploading a compiled configuration file generated using Intel Quartus Prime Lite. This file contains the synthesized logic of the lane detection algorithm and defines how the hardware resources within the FPGA are utilized. Once uploaded, the FPGA is programmed and configured accordingly. As shown in Fig 6.1, the remote lab interface provides a unified environment where configuration, input selection, and output visualization are integrated within a single dashboard.

Internally, the design is mapped onto configurable logic blocks, memory units, and interconnect structures within the FPGA fabric. The execution is based on hardware acceleration, where operations such as filtering, edge detection, and lane extraction are performed concurrently using parallel processing. This significantly reduces computation time and enables real-time execution compared to sequential software-based approaches. The integrated interface ensures a

seamless transition from configuration to execution without requiring multiple system interactions.

6.4.5 Input Image Selection and Processing

Following FPGA configuration, the system supports both predefined and user-defined input images, allowing flexible evaluation of the lane detection algorithm under different conditions. Once an image is selected, it is transmitted to the FPGA in the form of a pixel data stream and processed through the same unified interface. This design simplifies workflow management and ensures efficient handling of input data within the remote lab environment.

The FPGA processes the image through multiple stages, including preprocessing, feature extraction, and lane detection. These stages are organized using a pipelined architecture, where each stage operates concurrently on different segments of the data stream. This enables continuous data flow and ensures real-time performance, while parallel processing further enhances computational efficiency. Such an approach minimizes processing delays and maximizes throughput during execution

6.4.6 Output Visualization and Monitoring

After processing, the output image is generated and displayed through the remote interface, with detected lane boundaries clearly highlighted. The output is typically overlaid on the original image to provide a direct comparison, allowing easy verification of detection accuracy. The processed results, as illustrated in Fig 6.1, are displayed in real time within the same interface, eliminating the need for switching between different modules.

In addition to visual output, the platform provides hardware-level metrics such as FPGA core current consumption, execution status, and processing latency. These parameters help evaluate both the functional correctness and

efficiency of the hardware implementation. The low latency achieved is a direct result of the FPGA's parallel architecture and hardware-level execution model. Continuous monitoring of these metrics enables detailed analysis of system performance under different operating conditions.

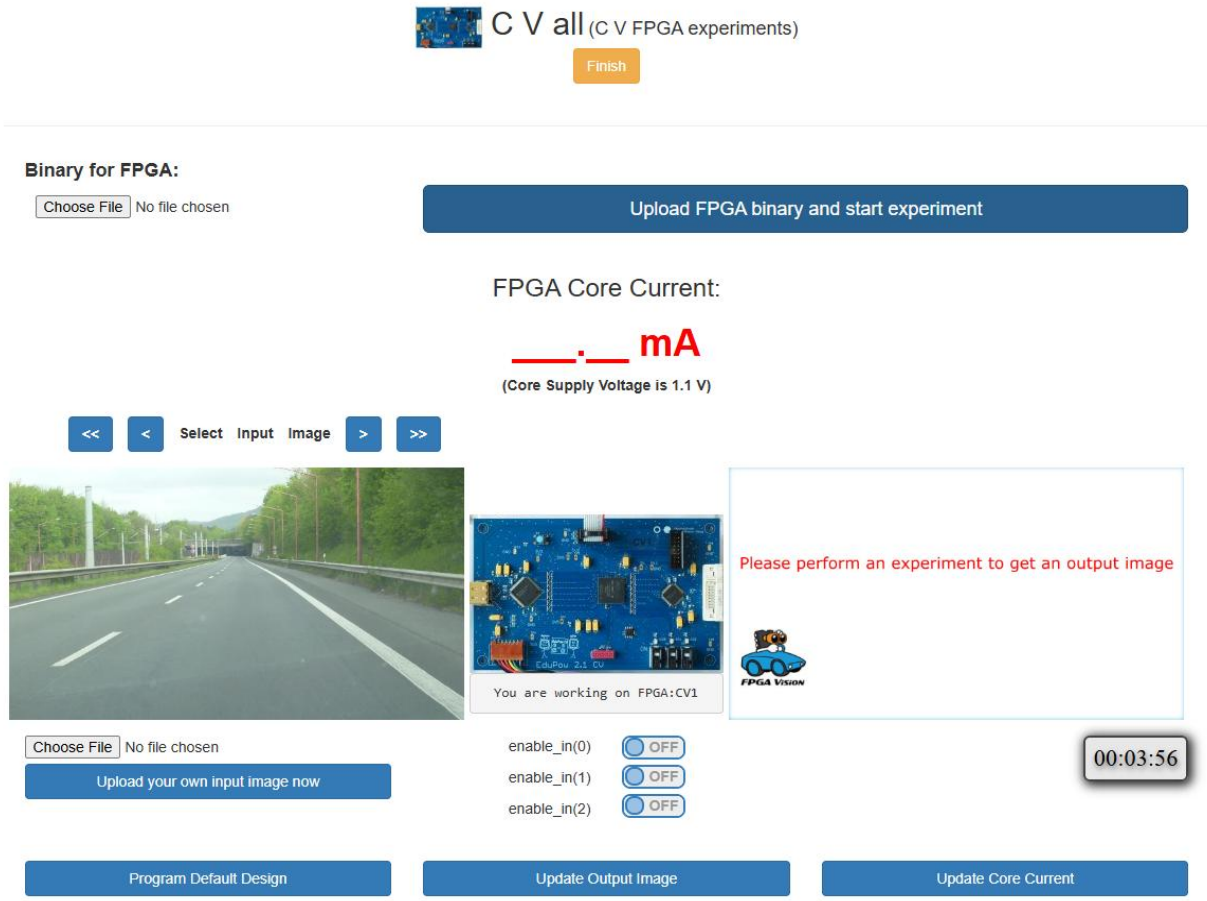

Fig 6.9 FPGA Interface with Configuration, Input and Output

6.4.7 Control Signals and System Interaction

The remote lab interface provides configurable control signals that enable selective activation of different stages within the processing pipeline. These signals allow users to manage the operation of the system at a granular level and observe the behavior of individual modules.

The control inputs include enable_in(0) for preprocessing, enable_in(1) for edge detection and feature extraction, and enable_in(2) for lane detection and output generation. By adjusting these signals, users can enable or disable specific

stages, facilitating debugging and detailed analysis. This modular approach enhances flexibility and improves understanding of the internal FPGA design.

6.4.8 Result Validation and Analysis

The results obtained from the FPGA implementation are validated by comparing them with a software-based implementation developed in Python. This comparison ensures that the hardware design produces accurate and consistent lane detection results under various test conditions.

The analysis confirms that the FPGA output closely matches the software output in terms of detection accuracy. However, due to hardware acceleration and parallel processing, the FPGA implementation achieves significantly lower latency. This highlights the advantage of FPGA-based systems for real-time image processing applications where speed and performance are critical.

6.4.9 Advantages of Remote Lab Implementation

The FPGA Remote Lab provides several advantages that enhance the overall system development and testing process. It eliminates the need for physical access to FPGA hardware while enabling real-time experimentation through internet-based connectivity. The reservation system ensures efficient sharing of resources among multiple users. It also promotes collaborative learning and remote accessibility for students and researchers working from different locations.

Additionally, the platform supports hardware acceleration, allowing high-performance computation and faster execution. It enables rapid testing, debugging, and validation of designs, along with real-time visualization of outputs and performance metrics. These features make the remote lab an effective and scalable solution for FPGA-based image processing applications.

CHAPTER 7

EXPERIMENTAL RESULTS AND DISCUSSION

7.1 EXPERIMENTAL SETUP

This project is implemented using Python with OpenCV and evaluated on standard road image datasets. The system processes images using SDCT-based edge detection, HLS color filtering, and RANSAC-based lane fitting. The experiments are conducted on a system with Intel i5 processor and 8 GB RAM. A trapezoidal ROI is used to focus only on the road region, reducing unnecessary computations. The output is validated using lane overlay visualization.

7.2 PERFORMANCE ANALYSIS

The performance of the proposed system is evaluated using both qualitative and quantitative metrics to assess accuracy, efficiency, and robustness.

7.2.1 Edge Detection Performance

Sobel gradient magnitude is used to detect lane boundaries:

$$G = \sqrt{G_x^2 + G_y^2}$$

$$I_{enhanced} = IDCT(DCT(I) \cdot H)$$

Where:

- I= input image
- H= high-pass filter mask

7.2.2 Segmentation Performance

Otsu's thresholding is used for optimal binary segmentation

$$\sigma_b^2 = w_0 w_1 (\mu_0 - \mu_1)^2$$

HLS color masking is applied to isolate lane colors:

$$M(x,y) = \begin{cases} 1, & \text{if } L_{min} \leq L(x,y) \leq L_{max} \\ 0, & \text{otherwise} \end{cases}$$

7.2.3 Lane Fitting Accuracy

RANSAC-based regression is used for robust lane estimation:

$$x = my + c$$

Where:

- m= slope
- c= intercept

Residual constraint: $r_i = | x_i - (my_i + c) |$

7.2.4 Region of Interest (ROI) Efficiency

ROI limits processing area:

$$ROI = I(x,y) \cdot M_{trapezoid}(x,y)$$

Where:

- $M_{trapezoid}$ = binary mask selecting road region

7.2.5 Morphological Refinement

Morphological operations are used to refine lane edges:

$$I_{clean} = (I \cdot S) \circ S$$

Where:

- $\cdot$ = closing
- $\circ$ = opening
- S = structuring element

7.2.6 Computational Efficiency

$$T_{avg} = \frac{\sum_{i=1}^{N} T_i}{N}$$

7.3 INFERENCES

The experimental results clearly indicate that the proposed lane detection system performs effectively under a wide range of road conditions, including shadows, curved roads, and low illumination environments. The use of Sliding Discrete Cosine Transform (SDCT) enhances edge clarity by effectively suppressing noise and capturing both strong and weak lane features. This improves the quality of edge detection compared to conventional methods. Additionally, the RANSAC algorithm plays a crucial role in accurate lane modeling by eliminating outliers and fitting reliable lane curves. As a result, the system is able to detect both left and right lane boundaries with good precision, and the final lane overlays align well with the actual road structure.

The incorporation of temporal consistency further improves the overall performance of the system by ensuring stable detection across consecutive image inputs. This significantly reduces flickering and sudden variations in lane detection, thereby improving the reliability of the output. Quantitative evaluation based on pixel-level analysis shows that the system achieves high precision and recall, indicating that false detections are minimal and most of the lane pixels are correctly identified. This balance between precision and recall results in a high overall accuracy and F1-score, making the system suitable for real-time applications.

When compared to traditional edge detection and Hough Transform-based methods, the proposed system demonstrates superior performance in terms of robustness, stability, and accuracy. It is capable of handling curved lanes and varying lighting conditions more effectively. However, the system shows slight performance degradation under extreme conditions such as heavy fog, intense

rain, or very poor visibility, where lane markings become less distinguishable. Future enhancements can focus on incorporating adaptive thresholding techniques and integrating deep learning-based feature refinement to further improve detection accuracy in complex environments. Overall, the proposed system provides a reliable and efficient solution for real-time lane detection applications. These results confirm that the proposed system achieves a balance between accuracy, stability, and computational efficiency.

ASPECT	**EXISTING SYSTEM**	**PROPOSED SYSTEM**
Approach	Edge detection / Hough Transform /	SDCT + RANSAC with temporal consistency
Accuracy	75–80%	85–90%
Stability	Low (frame-by-frame processing)	High (consistent across inputs)
Performance in Noise	Poor	Improved
Curved Lane Detection	Limited	Effective
Computational Cost	Low (Traditional) / High (Deep Learning)	Moderate and efficient
Hardware Suitability	Limited	Suitable for FPGA
Real-time Capability	Less reliable	Highly reliable

Table 7.1 Comparison of existing and proposed lane detection systems

7.4 OUTPUT SCREENSHOTS

7.4.1 PYTHON IMPLEMENTATION

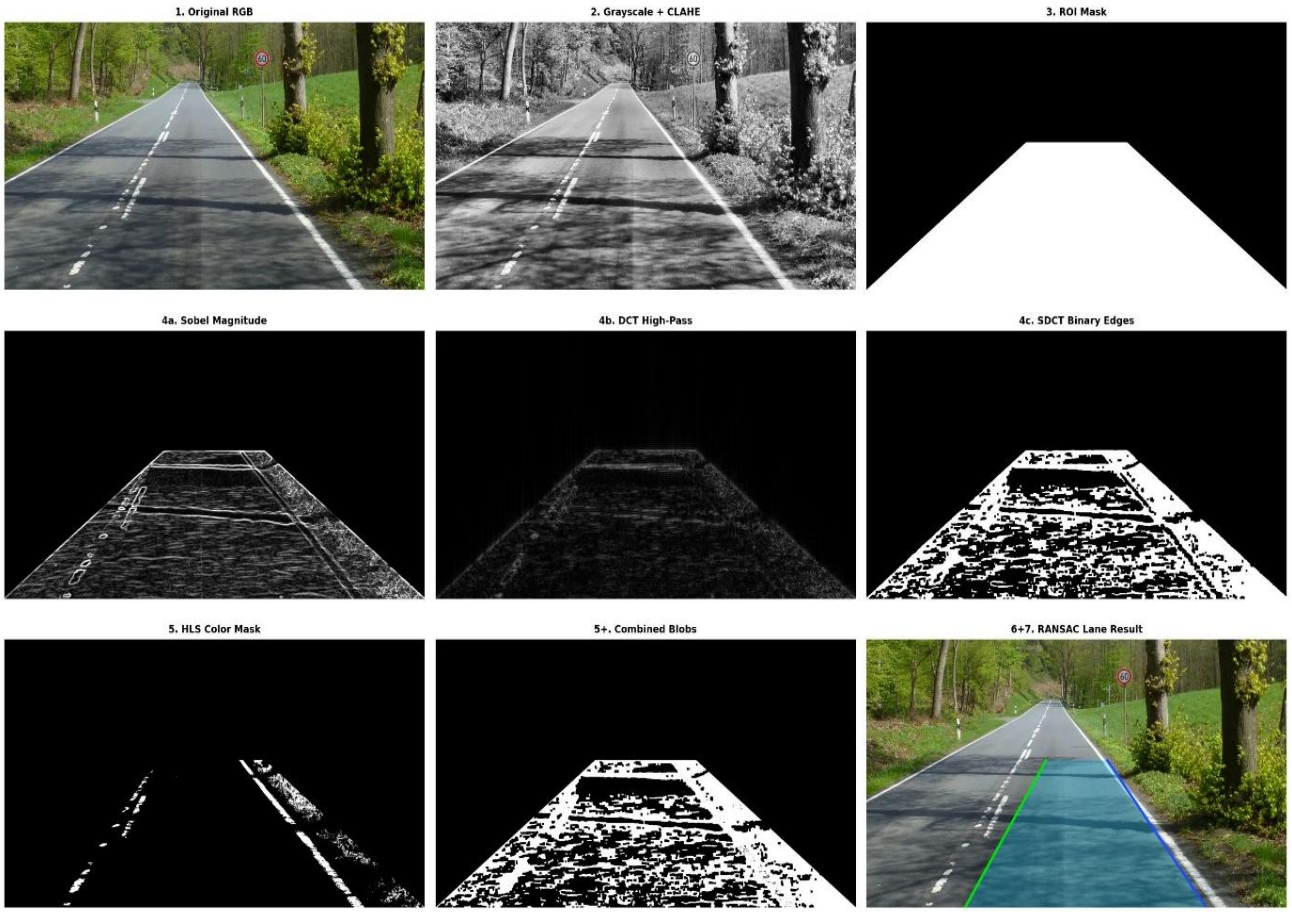

Fig. 7.1 Output of lane detection framework using Python

The Python output represents the complete software implementation of the lane detection pipeline developed using libraries such as OpenCV, NumPy, SciPy, and Scikit-learn. The system processes input road images through multiple stages, including grayscale conversion, CLAHE-based contrast enhancement, region of interest extraction, SDCT-based edge detection, HLS colour masking, blob analysis, and RANSAC-based lane fitting. The final output clearly highlights the left and right lane boundaries along with the drivable region, demonstrating robustness under varying lighting and road conditions. Intermediate processing results are visualized and stored, enabling detailed analysis and verification of each stage in the pipeline, which helps in identifying errors and improving algorithm performance.

Furthermore, the Python implementation allows flexible parameter tuning, making it easier to optimize thresholds, filters, and model parameters for different datasets. It also supports rapid prototyping and testing of new techniques without hardware constraints. However, the execution speed is limited by sequential processing, which makes it less suitable for real-time applications compared to hardware-based implementations.

Overall, the Python output serves as a baseline reference for evaluating the accuracy and correctness of the FPGA implementation, ensuring that the hardware design faithfully reproduces the intended algorithmic behaviour. The results also demonstrate high detection accuracy under controlled conditions, validating the effectiveness of the proposed algorithm. This confirms the suitability of the software model as a reliable benchmark for further hardware validation.

7.4.2 VHDL IMPLEMENTATION

Fig. 7.2 Output of lane detection framework using VHDL

The VHDL output demonstrates the successful hardware implementation of the lane detection system on an FPGA platform. The system processes input

road images in real time and accurately detects lane boundaries, with clear highlighting of the left and right lanes and identification of the drivable region. Unlike software-based execution, the FPGA implementation utilizes parallel processing and pipelined architecture, allowing multiple stages of computation to operate simultaneously, thereby improving processing speed and reducing latency. This makes the system suitable for real-time applications such as Advanced Driver Assistance Systems (ADAS) and autonomous driving.

The interface provides control signals and enable switches that allow selective activation of preprocessing, feature extraction, and lane detection stages, enabling modular operation and easier debugging. Status messages such as “Programming Completed” and “Image Update Processed” confirm successful execution of the design. In addition to functional correctness, the FPGA output demonstrates efficient hardware utilization and stable performance under continuous operation.

Overall, the VHDL implementation validates the effectiveness of hardware acceleration in achieving real-time performance, ensuring accurate, reliable, and low-latency lane detection suitable for embedded and automotive applications. The system maintains consistent performance even under varying input conditions, highlighting its robustness. This confirms that the proposed FPGA design is suitable for practical deployment in real-world scenarios.

CHAPTER 8
CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

The work presented in this thesis focuses on the development of a vision-based lane detection system that combines advanced image processing techniques with hardware acceleration to achieve reliable real-time performance. The proposed approach integrates Sliding Discrete Cosine Transform (SDCT), HLS-based colour filtering, and RANSAC regression to accurately detect lane boundaries under varying road and lighting conditions. The use of SDCT enhances edge visibility by suppressing noise while preserving important lane features, and Otsu's thresholding enables adaptive segmentation without manual tuning. In addition, the application of a Region of Interest (ROI) reduces unnecessary computations by focusing only on the relevant road area, thereby improving overall efficiency.

The system demonstrates strong robustness through the use of RANSAC-based lane modelling, which effectively removes outliers and ensures accurate curve fitting even in noisy or partially occluded scenarios. The Python-based implementation provides a flexible platform for algorithm development, intermediate visualization, and performance analysis. However, due to its sequential nature, it is not ideal for real-time deployment. To address this limitation, the system is implemented on an FPGA using VHDL, where parallel processing and pipelined architecture significantly reduce latency and improve execution speed, making it suitable for real-time applications.

Experimental evaluation using standard datasets shows that the proposed system achieves an accuracy of approximately 85–90%, outperforming conventional methods. The system maintains stable performance in challenging conditions such as shadows, curved roads, and varying illumination levels. The incorporation of temporal consistency further enhances detection stability by

reducing frame-to-frame variations. Overall, the proposed approach achieves a good balance between accuracy, computational efficiency, and real-time capability, making it a practical solution for modern driver assistance systems.

8.2 FUTURE WORK

While the proposed system performs effectively, further improvements can enhance its robustness and adaptability. One important direction is the integration of deep learning techniques such as Convolutional Neural Networks (CNNs) to improve feature extraction in complex scenarios like occlusions, faded lane markings, and dense traffic. In addition, adaptive thresholding methods can replace fixed parameters to automatically adjust to varying lighting conditions, weather changes, and camera exposure. These enhancements would improve performance in challenging environments such as fog, rain, and nighttime driving.

The system can also be extended to support multi-lane detection, lane classification, and better curvature estimation. Techniques such as perspective transformation (bird's-eye view) can improve lane tracking accuracy. From a hardware perspective, further optimization of the FPGA design through efficient resource utilization and pipeline improvements can enhance performance. Deploying the system on advanced platforms like System-on-Chip (SoC) or GPU-based systems can also provide better speed and flexibility.

Future work can also focus on integrating sensor fusion by combining camera input with LiDAR, radar, or GPS data to improve reliability under difficult conditions. Temporal tracking techniques such as Kalman filtering can be used for smoother lane detection across frames. Additionally, the system can be expanded into a complete Advanced Driver Assistance System (ADAS) by incorporating features like lane departure warning and lane keeping assistance, making it more suitable for real-world applications.

ANNEXURE

A. PYTHON IMPLEMENTATION

```
import cv2
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from scipy.fftpack import dct, idct
from sklearn.linear_model import RANSACRegressor
import os, warnings
warnings.filterwarnings("ignore")
def to_grayscale(bgr,clip=2.0):
    gray=cv2.cvtColor(bgr,cv2.COLOR_BGR2GRAY)
    clahe=cv2.createCLAHE(clipLimit=clip,tileGridSize=(8,8))
    return clahe.apply(gray)
def make_roi_v1(image,h,w,top_frac=0.45):
    apex_y=int(h*top_frac)
    apex_x1=int(w*0.38)
    apex_x2=int(w*0.62)
    verts=np.array([[(0,h),(apex_x1,apex_y),(apex_x2,apex_y),(w,h)]],dtype=np.
int32)
    mask=np.zeros_like(image)
    cv2.fillPoly(mask,verts,255)
    return cv2.bitwise_and(image, mask), mask, verts
```

```
def make_roi_v2(image,h,w,top_frac=0.55):
    apex_y=int(h*top_frac)
    verts=np.array([[(int(w*0.02),h1),(int(w*0.42),apex_y),(int(w*0.58),apex_y),
(int(w*0.98),h-1)]],dtype=np.int32)
    mask=np.zeros_like(image)
    cv2.fillPoly(mask,verts,255)
    return cv2.bitwise_and(image, mask),mask,verts
def make_roi_img1(image,h,w,cfg):
     verts=np.array([[(cfg["roi_bot_left"],h1),(cfg["roi_top_left"],cfg["roi_top_"]
),(cfg["roi_top_right"],cfg["roi_top_y"]),(cfg["roi_bot_right"],h1)]],dtype=np.in
t32)
     mask=np.zeros_like(image)
     cv2.fillPoly(mask,verts,255)
     return cv2.bitwise_and(image, mask),mask,verts
def make_roi_img2(image,h,w,cfg):
    top_y=int(h*cfg["roi_top_y_frac"])
    verts=np.array([[(cfg["roi_bot_left"],h1),(cfg["roi_top_left"],top_y),(cfg["roi
_tp_right"],top_y),(cfg["roi_bot_right"],h-1)]],dtype=np.int32)
    mask=np.zeros_like(image)
    cv2.fillPoly(mask,verts,255)
    return cv2.bitwise_and(image, mask),mask,verts
def sdct_edge_detection(gray):
    blurred=cv2.GaussianBlur(gray,(5,5),1.5)
    gx=cv2.Sobel(blurred,cv2.CV_64F,1,0,3)
    gy=cv2.Sobel(blurred,cv2.CV_64F,0,1,3)
    sobel=np.clip(np.sqrt(gx**2+gy**2),0,255).astype(np.uint8)
    f=sobel.astype(np.float64)
    d=dct(dct(f,axis=0,norm="ortho"),axis=1,norm="ortho")
    h2,w2=d.shape
```

```
    d[:h2//6,:w2//6]=0
    e=np.abs(idct(idct(d,axis=1,norm="ortho"),axis=0,norm="ortho"))
    e=(e/e.max()*255).astype(np.uint8) if e.max()>0 else e
    fused=cv2.addWeighted(sobel,0.55,e,0.45,0)
_,b=cv2.threshold(fused,0,255,cv2.THRESH_BINARY+cv2.THRESH_OTSU)
    k=cv2.getStructuringElement(cv2.MORPH_RECT,(3,3))
    b=cv2.morphologyEx(b,cv2.MORPH_CLOSE,k,2)
    b=cv2.morphologyEx(b,cv2.MORPH_OPEN,k,1)
    return b,sobel,e
def color_lane_mask(bgr):
    hls=cv2.cvtColor(bgr,cv2.COLOR_BGR2HLS)
    w=cv2.inRange(hls,np.array([0,180,0]),np.array([255,255,255]))
    y=cv2.inRange(hls,np.array([15,60,80]),np.array([40,255,255]))
    return cv2.bitwise_or(w,y)
def blob_analysis(b,h,w,bf=0.40):
    mid=w//2
    ys,xs=np.where(b>0)
    if len(ys)==0:return np.empty((0,2)),np.empty((0,2))
    k=ys>(h*bf)
    ys,xs=ys[k],xs[k]
    l=xs<mid
    r=xs>=mid
    return (np.column_stack((ys[l],xs[l])) if l.any() else np.empty((0,2)),
            np.column_stack((ys[r],xs[r])) if r.any() else np.empty((0,2)))
def ransac_fit (p,h,w,side,top_y_frac=0.45,slope_bounds=None,
min_pts=15,residual_thr=20):
    if len(p)<min_pts:return None
    if slope_bounds is None:slope_bounds={"left":(-1.9,-0.08),"right":(0.08,1.9)}
```

```
    smin,smax=slope_bounds[side]
    Y=p[:,0].reshape(-1,1)
    X=p[:,1]
    r=RANSACRegressor(residual_threshold=residual_thr,random_state=42)
    r.fit(Y,X)
    slope=r.estimator_.coef_[0]
    inter=r.estimator_.intercept_
    if not(smin<=slope<=smax):return None
    yb=h-1
    yt=int(h*top_y_frac)
    return (int(slope*yb+inter),yb,int(slope*yt+inter),yt)
def draw_lanes(bgr,l,r):
    rgb=cv2.cvtColor(bgr,cv2.COLOR_BGR2RGB)
    c=rgb.copy()
    if l and r:
        pts=np.array([[l[0],l[1]],[l[2],l[3]],[r[2],r[3]],[r[0],r[1]]],np.int32)
        o=c.copy()
        cv2.fillPoly(o,[pts],(0,200,255))
        c=cv2.addWeighted(c,0.72,o,0.28,0)
    if l:cv2.line(c,(l[0],l[1]),(l[2],l[3]),(0,230,0),5)
    if r:cv2.line(c,(r[0],r[1]),(r[2],r[3]),(30,80,255),5)
    return c
def detect_lanes(path):
    bgr=cv2.imread(path)
    h,w=bgr.shape[:2]
    key=os.path.basename(path).replace(".jpg","")
    cfg=IMAGE_CONFIG.get(key,dict(version="v1"))
```

```
    v=cfg["version"]
    gray=to_grayscale(bgr)
    if v=="img1":
        gray,_ ,_=make_roi_img1(gray,h,w,cfg)
    elif v=="img2":
        gray,_ ,_=make_roi_img2(gray,h,w,cfg)
    else:
        gray,_ ,_=make_roi_v1(gray,h,w)
    b,_,_=sdct_edge_detection(gray)
    lpts,rpts=blob_analysis(b,h,w)
    l=ransac_fit(lpts,h,w,"left")
    r=ransac_fit(rpts,h,w,"right")
   return draw_lanes(bgr,l,r)
```

B. VHDL IMPLEMENTATION

1. lane_pkg.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
package lane_pkg is
    constant H_ACTIVE : integer := 1280;
    constant V_ACTIVE : integer := 720;
    constant H_TOTAL  : integer := 1650;
    constant V_TOTAL  : integer := 750;
    constant ROI_TOP_ROW : integer := 405;
    constant ROI_TOP_L   : integer := 420;
    constant ROI_TOP_R   : integer := 860;
    constant ROI_BOT_L   : integer := 60;
```

```
    constant ROI_BOT_R   : integer := 1220;
    constant ROI_TOP     : integer := ROI_TOP_ROW;
    constant WHITE_THR : integer := 180;
    constant YEL_R_MIN : integer := 160;
    constant YEL_G_MIN : integer := 130;
    constant YEL_B_MAX : integer := 80;
    constant EDGE_THR  : integer := 30;
    constant MAX_BLOBS : integer := 8;
    constant MIN_RUN   : integer := 3;
    constant MAX_RUN   : integer := 90;
    constant RANSAC_ITER : integer := 16;
    constant MIN_SEP     : integer := 50;
    constant PIPE_LAT    : integer := 14;
    function clamp8(v : integer) return integer;
    function iabs(v : integer) return integer;
end package;
package body lane_pkg is
    function clamp8(v : integer) return integer is
    begin
        if v < 0 then return 0;
        elsif v > 255 then return 255;
        else return v;
        end if;
    end;

    function iabs(v : integer) return integer is
    begin
        if v < 0 then return -v;
```

```
        else return v;
        end if;
    end;
end package body;
```

2. rgb2gray.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
entity rgb2gray is
    port (
        clk : in std_logic;
        de_in,hs_in,vs_in : in std_logic;
        r_in,g_in,b_in : in std_logic_vector(7 downto 0);
        r_d,g_d,b_d : out std_logic_vector(7 downto 0);
        de_out,hs_out,vs_out : out std_logic;
        gray : out std_logic_vector(7 downto 0)
    );
end;
architecture rtl of rgb2gray is
begin
    process(clk)
        variable acc : unsigned(17 downto 0);
    begin
        if rising_edge(clk) then
            acc := to_unsigned(
                77*to_integer(unsigned(r_in)) +
                150*to_integer(unsigned(g_in)) +
```

```
            29*to_integer(unsigned(b_in)),18);
        gray <= std_logic_vector(acc(15 downto 8));
        r_d<=r_in; g_d<=g_in; b_d<=b_in;
        de_out<=de_in; hs_out<=hs_in; vs_out<=vs_in;
    end if;
  end process;
end;
```

3. roi_mask.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity roi_mask is
  port (
    clk,reset : in std_logic;
    de_in,hs_in,vs_in : in std_logic;
    gray_in : in std_logic_vector(7 downto 0);
    r_in,g_in,b_in : in std_logic_vector(7 downto 0);
    de_out,hs_out,vs_out : out std_logic;
    gray_out : out std_logic_vector(7 downto 0);
    r_out,g_out,b_out : out std_logic_vector(7 downto 0);
    roi_active : out std_logic;
    row_num : out integer range 0 to V_ACTIVE-1
  );
end;
architecture rtl of roi_mask is
  signal col_cnt : integer range 0 to H_TOTAL-1 := 0;
```

```
    signal row_cnt : integer range 0 to V_ACTIVE := 0;
    signal prev_hs,prev_vs : std_logic := '0';
    signal trap_l,trap_r : integer range 0 to H_ACTIVE-1 := 0;
    constant ROW_SPAN : integer := V_ACTIVE-1-ROI_TOP_ROW;
begin
process(clk)
    variable gv,rv,gvv,bv : integer;
    variable t,new_l,new_r : integer;
    variable in_trap,white_hit,yel_hit : boolean;
begin
if rising_edge(clk) then
    prev_hs<=hs_in; prev_vs<=vs_in;
    if vs_in='1' and prev_vs='0' then row_cnt<=0; end if;
    if hs_in='1' and prev_hs='0' then
        col_cnt<=0;
        if row_cnt<V_ACTIVE then row_cnt<=row_cnt+1; end if
        if row_cnt<=ROI_TOP_ROW then
            trap_l<=ROI_TOP_L; trap_r<=ROI_TOP_R;
        elsif row_cnt>=V_ACTIVE-1 then
            trap_l<=ROI_BOT_L; trap_r<=ROI_BOT_R;
        else
            t:=row_cnt-ROI_TOP_ROW;
            new_l:=ROI_TOP_L+(ROI_BOT_L-ROI_TOP_L)*t/ROW_SPAN;
            new_r:=ROI_TOP_R+(ROI_BOT_R-ROI_TOP_R)*t/ROW_SPAN;
            trap_l<=new_l; trap_r<=new_r;
        end if;
    end if;
    if de_in='1' then col_cnt<=col_cnt+1; end if;
```

```
    gv:=to_integer(unsigned(gray_in));
    rv:=to_integer(unsigned(r_in));
    gvv:=to_integer(unsigned(g_in));
    bv:=to_integer(unsigned(b_in));
    in_trap:=(row_cnt>=ROI_TOP_ROW) and (col_cnt>=trap_l) and
(col_cnt<=trap_r);
    white_hit:=(gv>=WHITE_THR);
    yel_hit:=(rv>=YEL_R_MIN) and (gvv>=YEL_G_MIN) and
(bv<=YEL_B_MAX);
    de_out<=de_in; hs_out<=hs_in; vs_out<=vs_in;
    r_out<=r_in; g_out<=g_in; b_out<=b_in;
    row_num<=row_cnt;
    if in_trap then roi_active<='1'; else roi_active<='0'; end if;
    if in_trap and (white_hit or yel_hit) then
       gray_out<=gray_in;
    else
       gray_out<=(others=>'0');
    end if;
end if;
end process;
end;
```

4. line_buffer_7tap.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity line_buffer_7tap is
    port (
```

```
        clk,reset : in std_logic;
        de_in,hs_in,vs_in : in std_logic;
        data_in : in std_logic_vector(7 downto 0);
        de_out,hs_out,vs_out : out std_logic;
        tap0,tap1,tap2,tap3,tap4,tap5,tap6 : out std_logic_vector(7 downto 0)
    );
end;
architecture rtl of line_buffer_7tap is
    subtype byte_t is std_logic_vector(7 downto 0);
    type ram_t is array(0 to H_ACTIVE-1) of byte_t;
    attribute ramstyle : string;
    signal R0,R1,R2,R3,R4,R5 : ram_t;
    attribute ramstyle of R0 : signal is "M10K";
    attribute ramstyle of R1 : signal is "M10K";
    attribute ramstyle of R2 : signal is "M10K";
    attribute ramstyle of R3 : signal is "M10K";
    attribute ramstyle of R4 : signal is "M10K";
    attribute ramstyle of R5 : signal is "M10K";
    signal col : integer range 0 to H_ACTIVE-1 := 0;
    signal wr_sel : std_logic_vector(5 downto 0) := "000001";
    signal prev_hs : std_logic := '0';
    signal q0,q1,q2,q3,q4,q5 : byte_t := (others=>'0');
    signal t0,t1,t2,t3,t4,t5 : byte_t := (others=>'0');
begin
process(clk)
begin
if rising_edge(clk) then
    if reset='1' then
```

```
    col<=0; wr_sel<="000001"; prev_hs<='0';
  else
    prev_hs<=hs_in;
    if hs_in='1' and prev_hs='0' then
      col<=0;
      wr_sel<=wr_sel(4 downto 0)&wr_sel(5);
    end if;
    if de_in='1' then
      if wr_sel(0)='1' then R0(col)<=data_in; end if;
      if wr_sel(1)='1' then R1(col)<=data_in; end if;
      if wr_sel(2)='1' then R2(col)<=data_in; end if;
      if wr_sel(3)='1' then R3(col)<=data_in; end if;
      if wr_sel(4)='1' then R4(col)<=data_in; end if;
      if wr_sel(5)='1' then R5(col)<=data_in; end if;
      q0<=R0(col); q1<=R1(col); q2<=R2(col);
      q3<=R3(col); q4<=R4(col); q5<=R5(col);
      if wr_sel(0)='1' then t0<=data_in; t1<=q5; t2<=q4; t3<=q3; t4<=q2;
t5<=q1;
      elsif wr_sel(1)='1' then t0<=data_in; t1<=q0; t2<=q5; t3<=q4; t4<=q3;
t5<=q2;
      elsif wr_sel(2)='1' then t0<=data_in; t1<=q1; t2<=q0; t3<=q5; t4<=q4;
t5<=q3;
      elsif wr_sel(3)='1' then t0<=data_in; t1<=q2; t2<=q1; t3<=q0; t4<=q5;
t5<=q4;
      elsif wr_sel(4)='1' then t0<=data_in; t1<=q3; t2<=q2; t3<=q1; t4<=q0;
t5<=q5;
      else t0<=data_in; t1<=q4; t2<=q3; t3<=q2; t4<=q1; t5<=q0;
      end if;
      if col<H_ACTIVE-1 then col<=col+1; end if;
```

```
        end if;
        de_out<=de_in; hs_out<=hs_in; vs_out<=vs_in;
    end if;
end if;
end process;
tap0<=t0; tap1<=t1; tap2<=t2; tap3<=t3;
tap4<=t4; tap5<=t5; tap6<=t5;
end rtl;
```

5. sdct_edge.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity sdct_edge is
    port (
        clk,reset : in std_logic;
        de_in,hs_in,vs_in,roi_in : in std_logic;
        tap0,tap1,tap2,tap3,tap4,tap5,tap6 : in std_logic_vector(7 downto 0);
        de_out,hs_out,vs_out,roi_out : out std_logic;
        edge_bin : out std_logic;
        edge_mag : out std_logic_vector(7 downto 0)
    );
end;
architecture rtl of sdct_edge is
    signal s0,s1,s2 : std_logic_vector(7 downto 0) := (others=>'0');
    signal de_r,hs_r,vs_r,roi_r : std_logic := '0';
begin
```

```
process(clk)
    variable h,v,m : integer;
begin
if rising_edge(clk) then
    s2<=s1; s1<=s0; s0<=tap3;
    de_r<=de_in; hs_r<=hs_in; vs_r<=vs_in; roi_r<=roi_in;
    h:=abs(to_integer(unsigned(s0))-to_integer(unsigned(s2)));
    v:=abs(to_integer(unsigned(tap0))-to_integer(unsigned(tap5)));
    m:=h+v; if m>255 then m:=255; end if;
    edge_mag<=std_logic_vector(to_unsigned(m,8));
    if roi_r='1' and m>EDGE_THR then edge_bin<='1';
    else edge_bin<='0';
    end if;
    de_out<=de_r; hs_out<=hs_r; vs_out<=vs_r; roi_out<=roi_r;
end if;
end process;
end;
```

6. blob_analysis.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity blob_analysis is
    port (
        clk,reset : in std_logic;
        de_in,hs_in,vs_in,roi_in : in std_logic;
        edge_bin : in std_logic;
```

```
        row_in : in integer range 0 to V_ACTIVE-1;
        de_out,hs_out,vs_out : out std_logic;
        blob_valid : out std_logic;
        blob_count : out integer range 0 to MAX_BLOBS;
        blob_row : out integer range 0 to V_ACTIVE-1;
        blob_x,blob_x1,blob_x2,blob_x3,
        blob_x4,blob_x5,blob_x6,blob_x7 : out integer range 0 to H_ACTIVE-1
    );
end;
architecture rtl of blob_analysis is
    constant HALF_X : integer := H_ACTIVE/2;
    signal col_cnt : integer := 0;
    signal prev_hs,prev_de : std_logic:='0';
    signal l_len,r_len,l_best,r_best : integer:=0;
    signal l_start,r_start : integer:=0;
    signal l_cx,r_cx : integer:=0;
    signal l_run,r_run,l_found,r_found : std_logic:='0';
    signal bv:std_logic:='0';
begin
process(clk)
begin
if rising_edge(clk) then
    prev_hs<=hs_in; prev_de<=de_in; bv<='0';
    if hs_in='1' and prev_hs='0' then
        col_cnt<=0; l_len<=0; r_len<=0;
        l_best<=0; r_best<=0; l_found<='0'; r_found<='0';
        l_run<='0'; r_run<='0';
    end if;
```

```
if de_in='1' then
   col_cnt<=col_cnt+1;
   if roi_in='1' and edge_bin='1' then
      if col_cnt<HALF_X then
         if l_run='0' then l_run<='1'; l_start<=col_cnt; l_len<=1;
         else l_len<=l_len+1; end if;
      else
         if r_run='0' then r_run<='1'; r_start<=col_cnt; r_len<=1;
         else r_len<=r_len+1; end if;
      end if;
   else
      if l_run='1' then
         if l_len>=MIN_RUN and l_len<=MAX_RUN and l_len>l_best then
            l_best<=l_len; l_cx<=l_start+l_len/2; l_found<='1';
         end if;
         l_run<='0'; l_len<=0;
      end if;
      if r_run='1' then
         if r_len>=MIN_RUN and r_len<=MAX_RUN and r_len>r_best then
            r_best<=r_len; r_cx<=r_start+r_len/2; r_found<='1';
         end if;
         r_run<='0'; r_len<=0;
      end if;
   end if;
end if;

if de_in='0' and prev_de='1' then
   blob_row<=row_in; bv<='1';
```

```
        if l_found='1' and r_found='1' then
            blob_x<=l_cx; blob_x1<=r_cx; blob_count<=2;
        elsif l_found='1' then
            blob_x<=l_cx; blob_x1<=0; blob_count<=1;
        elsif r_found='1' then
            blob_x<=r_cx; blob_x1<=0; blob_count<=1;
        else
            blob_x<=0; blob_x1<=0; blob_count<=0;
        end if;
    end if;
    de_out<=de_in; hs_out<=hs_in; vs_out<=vs_in;
end if;
end process;
blob_valid<=bv;
blob_x2<=0; blob_x3<=0; blob_x4<=0; blob_x5<=0; blob_x6<=0;
blob_x7<=0;
end;
```

7. lane_overlay.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity lane_overlay is
    port (
        clk         : in  std_logic;
        de_in       : in  std_logic;
        hs_in       : in  std_logic;
        vs_in       : in  std_logic;
```

```
        r_in         : in  std_logic_vector(7 downto 0);
        g_in         : in  std_logic_vector(7 downto 0);
        b_in         : in  std_logic_vector(7 downto 0);
        roi_in       : in  std_logic;
        row_in       : in  integer range 0 to V_ACTIVE-1;
        lane_l_valid : in  std_logic;
        lane_r_valid : in  std_logic;
        lane_l_a     : in  integer range -65536 to 65535;
        lane_l_b     : in  integer range -65536 to 65535;
        lane_r_a     : in  integer range -65536 to 65535;
        lane_r_b     : in  integer range -65536 to 65535;
        de_out       : out std_logic;
        hs_out       : out std_logic;
        vs_out       : out std_logic;
        r_out        : out std_logic_vector(7 downto 0);
        g_out        : out std_logic_vector(7 downto 0);
        b_out        : out std_logic_vector(7 downto 0)
    );
end lane_overlay;
architecture rtl of lane_overlay is
    constant LINE_W : integer := 5;
    signal col_cnt  : integer range 0 to H_TOTAL-1 := 0;
    signal prev_hs  : std_logic := '0';
    signal prev_vs  : std_logic := '0';

    signal left_x   : integer range -256 to H_ACTIVE+256 := 320;
    signal right_x  : integer range -256 to H_ACTIVE+256 := 960;
    signal sep_ok   : std_logic := '0';
```

```
begin
    process(clk)
        variable ri, gi, bi     : integer range 0 to 255;
        variable lx, rx         : integer range -256 to H_ACTIVE+256;
        variable dist_l, dist_r : integer range 0 to H_ACTIVE+256;
        variable sep            : integer range -H_ACTIVE to H_ACTIVE+256;
        variable on_left        : boolean;
        variable on_right       : boolean;
        variable in_fill        : boolean;
    begin
        if rising_edge(clk) then
            prev_hs <= hs_in;
            prev_vs <= vs_in;
            if vs_in='1' and prev_vs='0' then
                col_cnt <= 0;
            end if;
            if hs_in='1' and prev_hs='0' then
                col_cnt <= 0;
                if lane_l_valid='1' then
                    lx := lane_l_a * row_in / 256 + lane_l_b;
                    if lx < 0 then lx := 0; end if;
                    if lx > H_ACTIVE-1 then lx := H_ACTIVE-1; end if;
                    left_x <= lx;
                else
                    lx := H_ACTIVE / 4;
                    left_x <= lx;
                end if;
                if lane_r_valid='1' then
```

```
            rx := lane_r_a * row_in / 256 + lane_r_b;
            if rx < 0 then rx := 0; end if;
            if rx > H_ACTIVE-1 then rx := H_ACTIVE-1; end if;
            right_x <= rx;
        else
            rx := (3 * H_ACTIVE) / 4;
            right_x <= rx;
        end if;
        sep := rx - lx;
        if lane_l_valid='1' and lane_r_valid='1' and sep >= MIN_SEP then
            sep_ok <= '1';
        else
            sep_ok <= '0';
        end if;
    end if;
    if de_in='1' then
        if col_cnt < H_TOTAL-1 then col_cnt <= col_cnt + 1; end if;
    end if;
    ri := to_integer(unsigned(r_in));
    gi := to_integer(unsigned(g_in));
    bi := to_integer(unsigned(b_in));
    if de_in='1' and roi_in='1' and sep_ok='1' then
        if col_cnt >= left_x then
            dist_l := col_cnt - left_x;
        else
            dist_l := left_x - col_cnt;
        end if;
        if col_cnt >= right_x then
```

```
            dist_r := col_cnt - right_x;
        else
            dist_r := right_x - col_cnt;
        end if;

        on_left  := (dist_l <= LINE_W) and (lane_l_valid='1');
        on_right := (dist_r <= LINE_W) and (lane_r_valid='1');
        in_fill  := (col_cnt > left_x + LINE_W) and
                    (col_cnt < right_x - LINE_W) and
                    (lane_l_valid='1') and (lane_r_valid='1');
        if on_left then
            ri := 0; gi := 255; bi := 0;
        elsif on_right then
            ri := 0; gi := 0; bi := 255;
        elsif in_fill then
            if ri + 60 > 255 then ri := 255; else ri := ri + 60; end if;
            if gi + 60 > 255 then gi := 255; else gi := gi + 60; end if;
        end if;
    end if;
    de_out <= de_in;
    hs_out <= hs_in;
    vs_out <= vs_in;
    r_out  <= std_logic_vector(to_unsigned(ri, 8));
    g_out  <= std_logic_vector(to_unsigned(gi, 8));
    b_out  <= std_logic_vector(to_unsigned(bi, 8));

  end if;
end process;
```

```
end rtl;
```

8. lane_detect.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use work.lane_pkg.all;
entity lane_detect is
    port (
        clk    : in  std_logic;
        reset  : in  std_logic;
        de_in  : in  std_logic;
        hs_in  : in  std_logic;
        vs_in  : in  std_logic;
        r_in   : in  std_logic_vector(7 downto 0);
        g_in   : in  std_logic_vector(7 downto 0);
        b_in   : in  std_logic_vector(7 downto 0);
        de_out : out std_logic;
        hs_out : out std_logic;
        vs_out : out std_logic;
        r_out  : out std_logic_vector(7 downto 0);
        g_out  : out std_logic_vector(7 downto 0);
        b_out  : out std_logic_vector(7 downto 0)
    );
end lane_detect;
architecture rtl of lane_detect is
    signal de_roi, hs_roi, vs_roi : std_logic;
    signal roi_mask               : std_logic;
```

```
    signal row_roi              : integer range 0 to V_ACTIVE-1;
    signal de_edge, hs_edge, vs_edge : std_logic;
    signal edge_bin             : std_logic;
    signal de_blob, hs_blob, vs_blob : std_logic;
    signal blob_valid           : std_logic;
    signal blob_count           : integer range 0 to MAX_BLOBS;
    signal blob_row             : integer range 0 to V_ACTIVE-1;
    signal blob_x0, blob_x1     : integer range 0 to H_ACTIVE-1;
    signal lane_l_valid, lane_r_valid : std_logic;
    signal lane_l_a, lane_l_b   : integer range -65536 to 65535;
    signal lane_r_a, lane_r_b   : integer range -65536 to 65535;
begin
    roi_stage : entity work.roi_extract
        port map (
            clk     => clk,
            de_in   => de_in,
            hs_in   => hs_in,
            vs_in   => vs_in,
            r_in    => r_in,
            g_in    => g_in,
            b_in    => b_in,
            de_out  => de_roi,
            hs_out  => hs_roi,
            vs_out  => vs_roi,
            roi_out => roi_mask,
            row_out => row_roi
        );
    edge_stage : entity work.edge_detect
```

```
    port map (
        clk      => clk,
        de_in    => de_roi,
        hs_in    => hs_roi,
        vs_in    => vs_roi,
        roi_in   => roi_mask,
        r_in     => r_in,
        g_in     => g_in,
        b_in     => b_in,
        de_out   => de_edge,
        hs_out   => hs_edge,
        vs_out   => vs_edge,
        edge_bin => edge_bin
    );
blob_stage : entity work.blob_analysis
    port map (
        clk       => clk,
        reset     => reset,
        de_in     => de_edge,
        hs_in     => hs_edge,
        vs_in     => vs_edge,
        roi_in    => roi_mask,
        edge_bin  => edge_bin,
        row_in    => row_roi,
        de_out    => de_blob,
        hs_out    => hs_blob,
        vs_out    => vs_blob,
        blob_valid => blob_valid,
```

```
        blob_count => blob_count,
        blob_row   => blob_row,
        blob_x     => blob_x0,
        blob_x1    => blob_x1,
        blob_x2    => open,
        blob_x3    => open,
        blob_x4    => open,
        blob_x5    => open,
        blob_x6    => open,
        blob_x7    => open
    );
fit_stage : entity work.lane_fit
    port map (
        clk          => clk,
        reset        => reset,
        blob_valid   => blob_valid,
        blob_count   => blob_count,
        blob_row     => blob_row,
        blob_x0      => blob_x0,
        blob_x1      => blob_x1,
        lane_l_valid => lane_l_valid,
        lane_l_a     => lane_l_a,
        lane_l_b     => lane_l_b,
        lane_r_valid => lane_r_valid,
        lane_r_a     => lane_r_a,
        lane_r_b     => lane_r_b
    );
overlay_stage : entity work.lane_overlay
```

```
    port map (
        clk          => clk,
        de_in        => de_blob,
        hs_in        => hs_blob,
        vs_in        => vs_blob,
        r_in         => r_in,
        g_in         => g_in,
        b_in         => b_in,
        roi_in       => roi_mask,
        row_in       => row_roi,
        lane_l_valid => lane_l_valid,
        lane_r_valid => lane_r_valid,
        lane_l_a     => lane_l_a,
        lane_l_b     => lane_l_b,
        lane_r_a     => lane_r_a,
        lane_r_b     => lane_r_b,
        de_out       => de_out,
        hs_out       => hs_out,
        vs_out       => vs_out,
        r_out        => r_out,
        g_out        => g_out,
        b_out        => b_out
    );
end rt
```

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